



Mechanisms underlying Cortical Variability Dynamics

Empirical Data Analysis and Spiking Attractor Model

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www.computational-systems-neuroscience.de



Mechanisms underlying Cortical Variability Dynamics

Empirical Data Analysis and Spiking Attractor Model

Motivation of Research Question

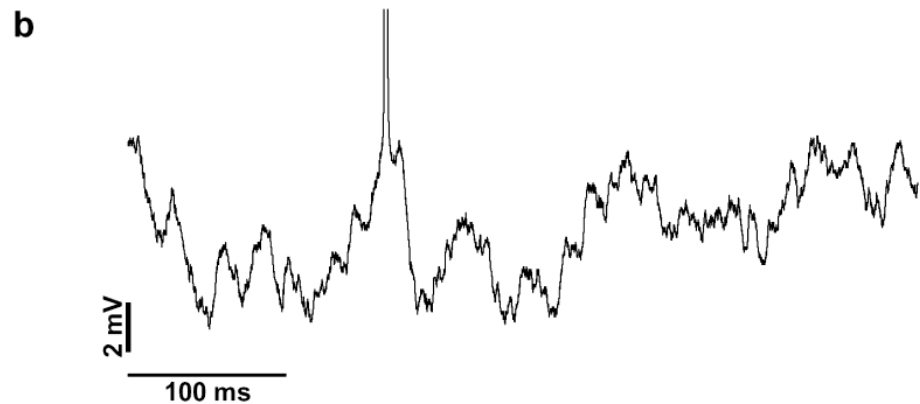
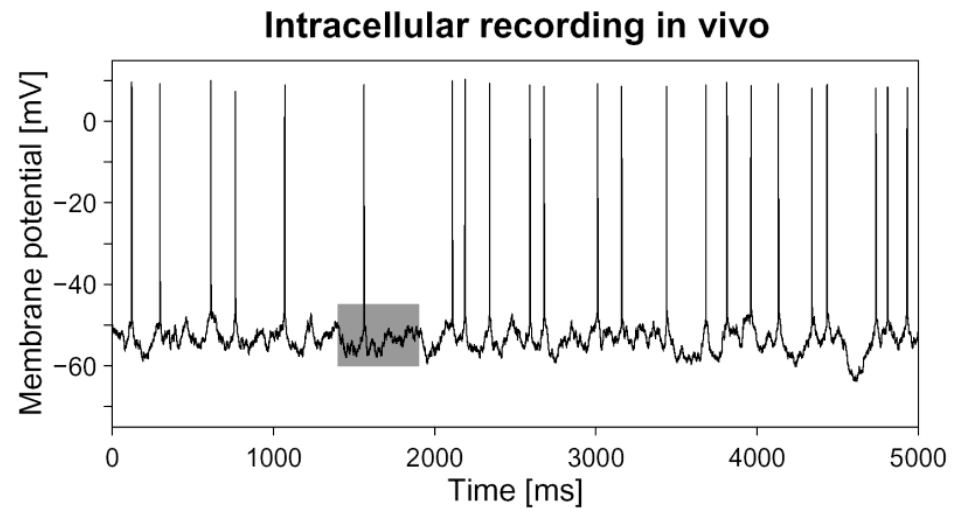
Part I - Empirical Analysis of Variability in Neuronal Spike Trains

Part II - Spiking Attractor Model Explains Cortical Variability Dynamics

Motivation of Research Question

Motivation

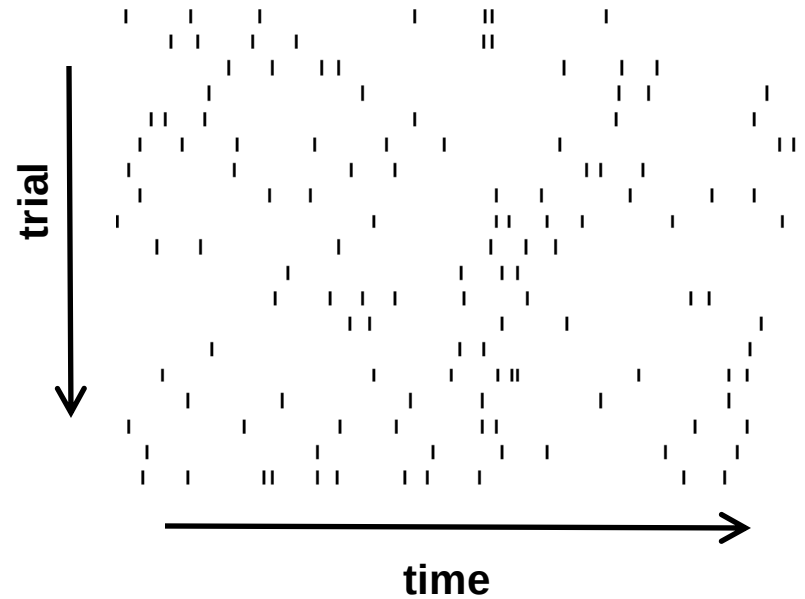
neocortex *in vivo* is **permanently active**



Leger, Stern, Aertsen, Heck (2004) *J Neurophysiol* 93

Motivation

neocortex *in vivo* is **permanently active**
spikes occur at **irregular intervals** in time

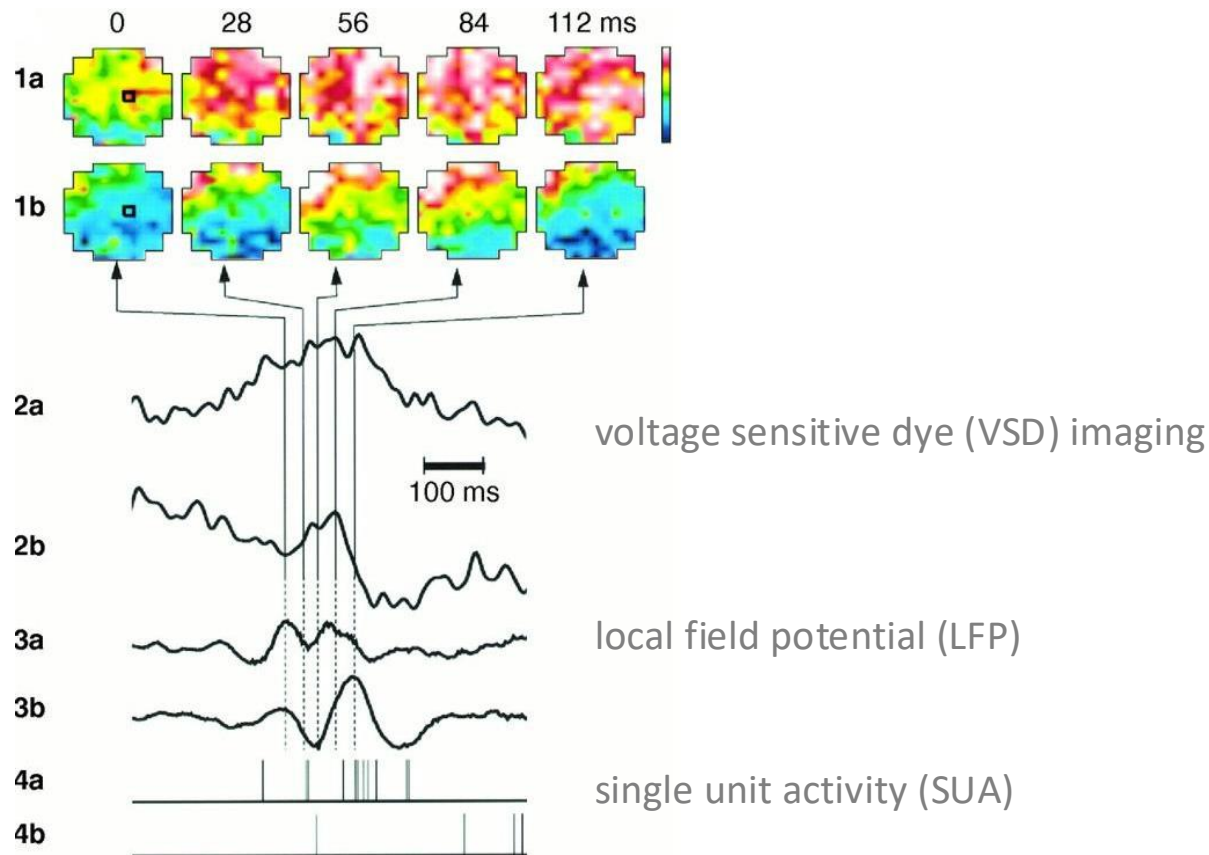


Motivation

neocortex *in vivo* is **permanently active**

spikes occur at **irregular intervals** in time

neural responses are **highly variable** upon repeated stimulation



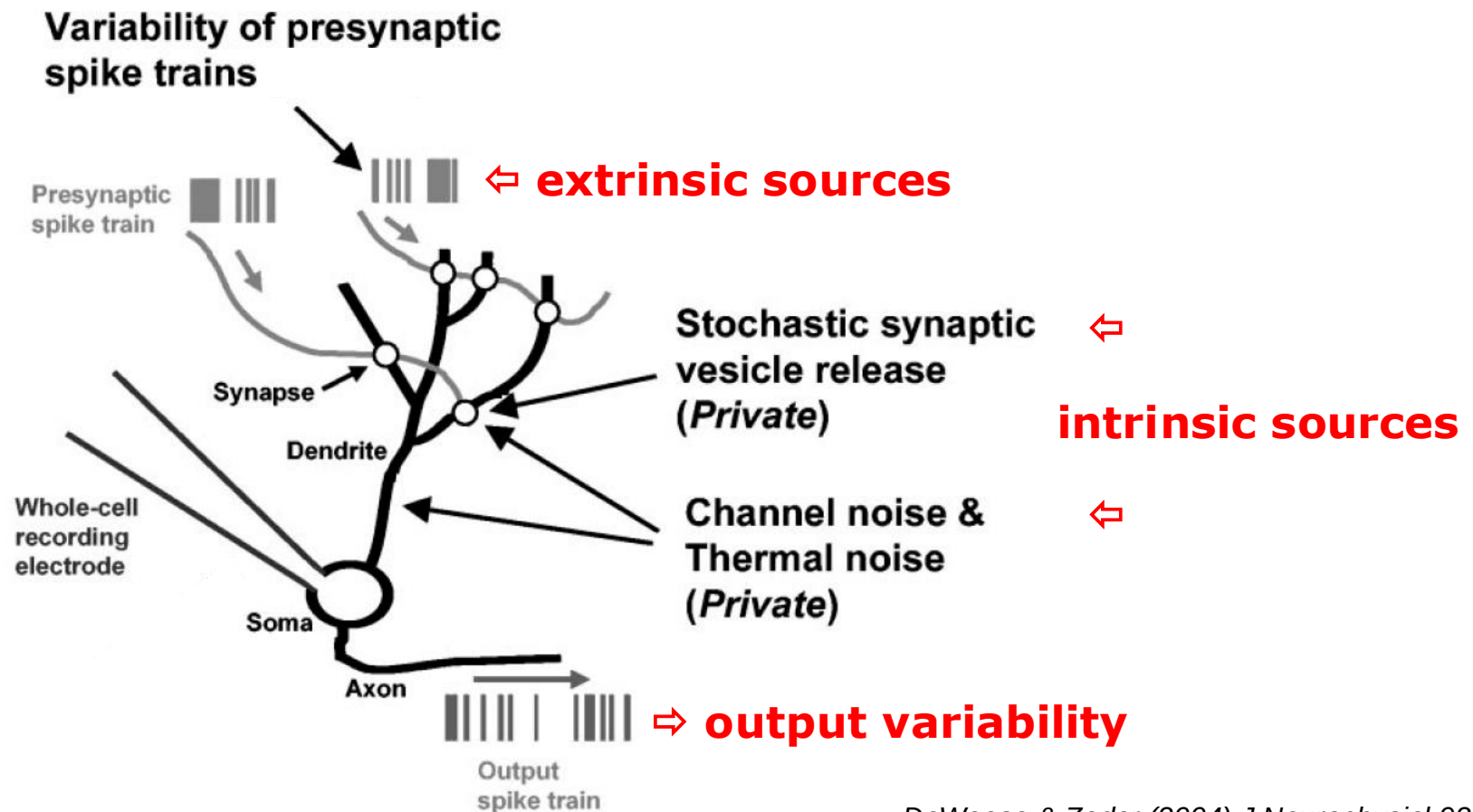
Motivation

neocortex *in vivo* is **permanently active**

spikes occur at **irregular intervals** in time

neural responses are **highly variable** upon repeated stimulation

intrinsic versus extrinsic sources of **single neuron variability**



The Variable Discharge of Cortical Neurons: Implications for Connectivity, Computation, and Information Coding

Michael N. Shadlen¹ and William T. Newsome²

3872 J. Neurosci., May 15, 1998, 18(10):3870–3896

Table 1. Properties of statistical homogeneity for cortical neurons

Response property	Approximate value
Dynamic range of response	Approximately 100–200 spikes/s
Distribution of interspike intervals	Approximately exponential
Spike count variance	Approximately 1.5 times the mean count
Spike rate modulation	Expected rate can vary from 1–15%

Fano Factor > 1

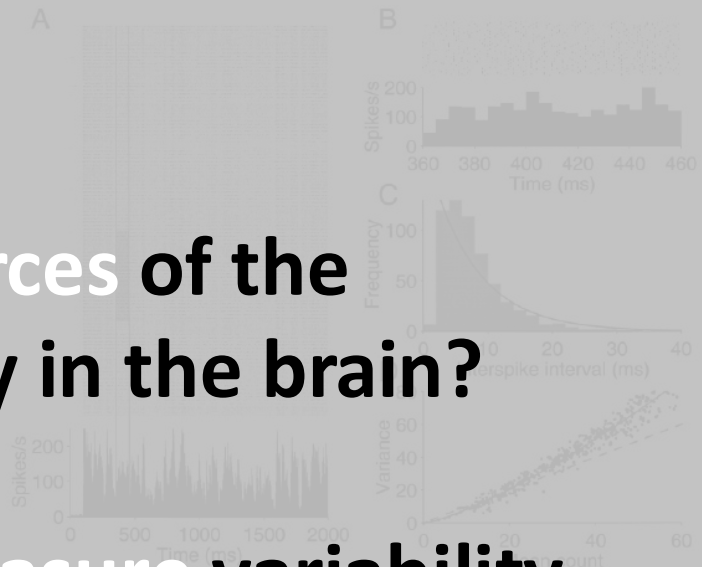
How can we reliably measure variability in our single neuron data?

Hypothesis :

⇒ single neuron spiking is a random Poisson process

⇒ **BUT : this is not true**

(and the Poisson model is a deficient model for neuronal spiking anyway)



Part I

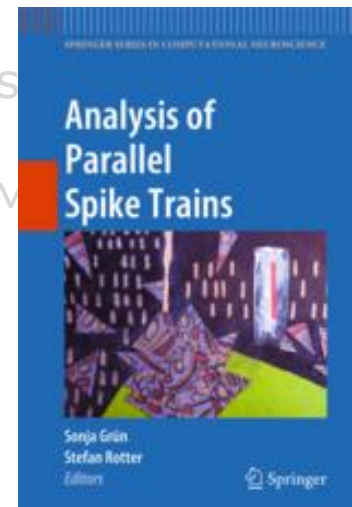
Empirical Analysis of Variability in Neuronal Spike Trains

- I Measuring Interval and Count Variability
- II Cortical Variability under Stationary Conditions
- III Cortical Variability Dynamics in the Behaving Monkey

Part I

Empirical Analysis of Variability in Neuronal Spike Trains

- I Measuring Interval and Count Variability
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- III Cortical Variability Dynamics in the Behaving Mouse



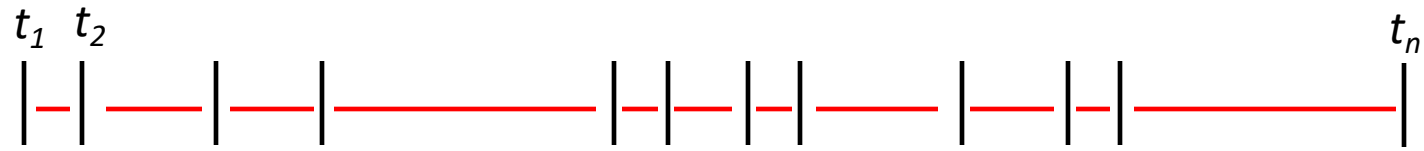
Grün, Rotter (Eds.), *Springer Series Comp Neurosci* 7 (2010)
Ch. 3. Analysis and Interpretation of Interval and Count Variability in Neural Spike Trains

Interval and Count: Random Variables

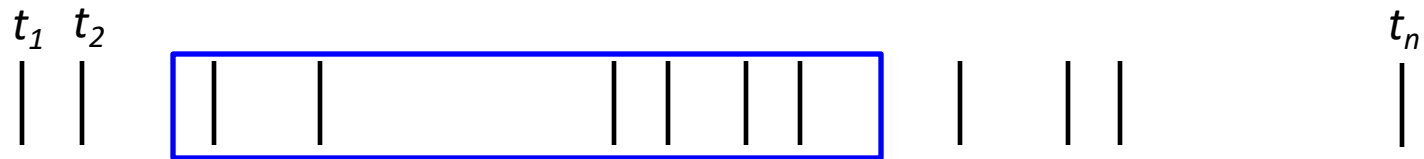


X_i = *Inter Spike Interval (ISI), continuous random variable*

Interval and Count: Random Variables

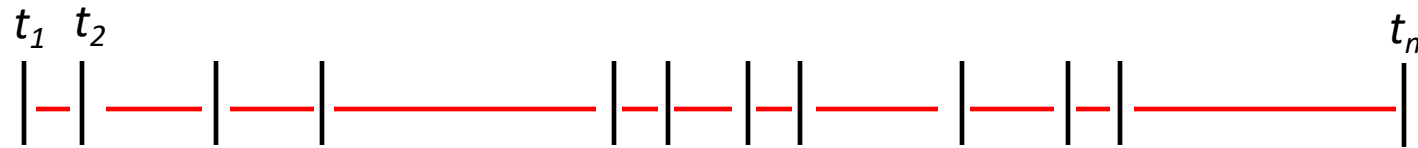


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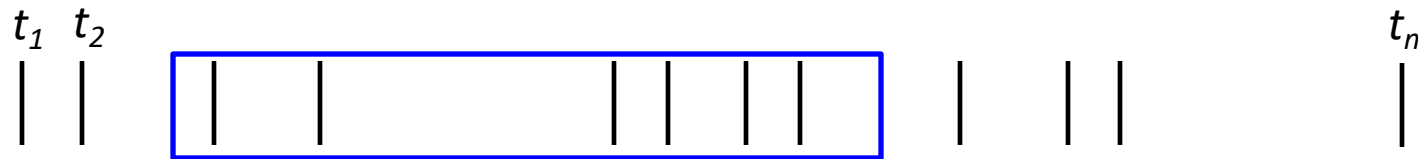


N_i = Spike Count, discrete random variable

Interval and Count: Random Variables



$\mathbf{X}_i$ = Inter Spike Interval (ISI), continuous random variable



$\mathbf{N}_i$ = Spike Count, discrete random variable

$\mathbf{N}_i$ and $\mathbf{X}_i$ are related for any given point process:

$$P\{\tau_k \leq t\} = P\{N_{[0,t)} \geq k\} \quad \tau_k = \sum_{i=1}^k X_i$$

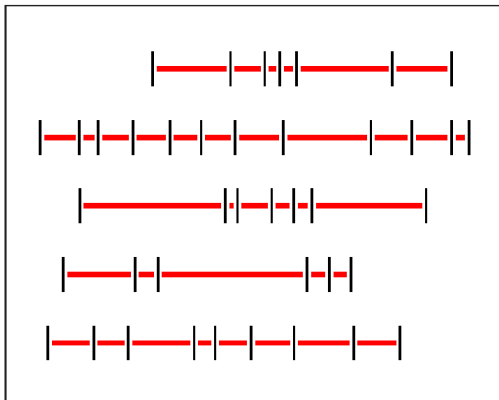
Assumptions : stationary and orderly point process

Interval and Count Variability: Expectation

Coefficient of variation

(of inter-spike intervals X)

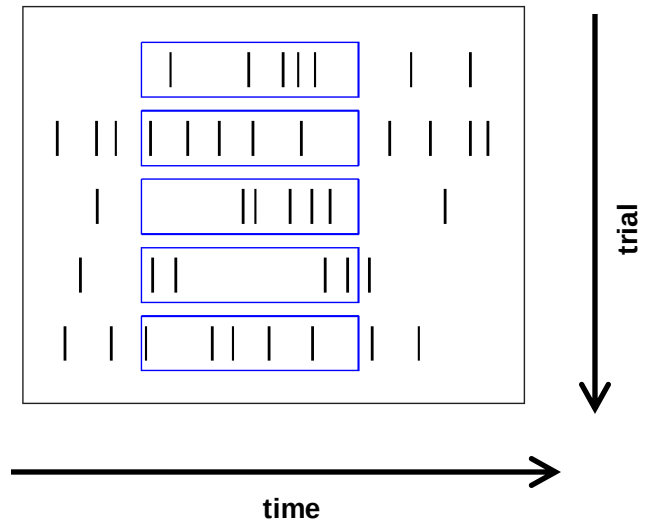
$$CV^2 = \frac{Var(ISI)}{mean^2(ISI)}$$



Fano factor

(of spike count N)

$$FF = \frac{Var(count)}{mean(count)}$$



Interval and Count Variability: Expectation

Coefficient of variation

(of inter-spike intervals X)

$$CV^2 = \frac{Var(ISI)}{mean^2(ISI)}$$

Fano factor

(of spike count N)

$$FF = \frac{Var(count)}{mean(count)}$$

$$\lim_{T \rightarrow \infty} FF = CV_{\infty}^2 (1 + 2\xi) \quad \text{with } \xi = \sum_{i=1}^{\infty} \xi_i$$

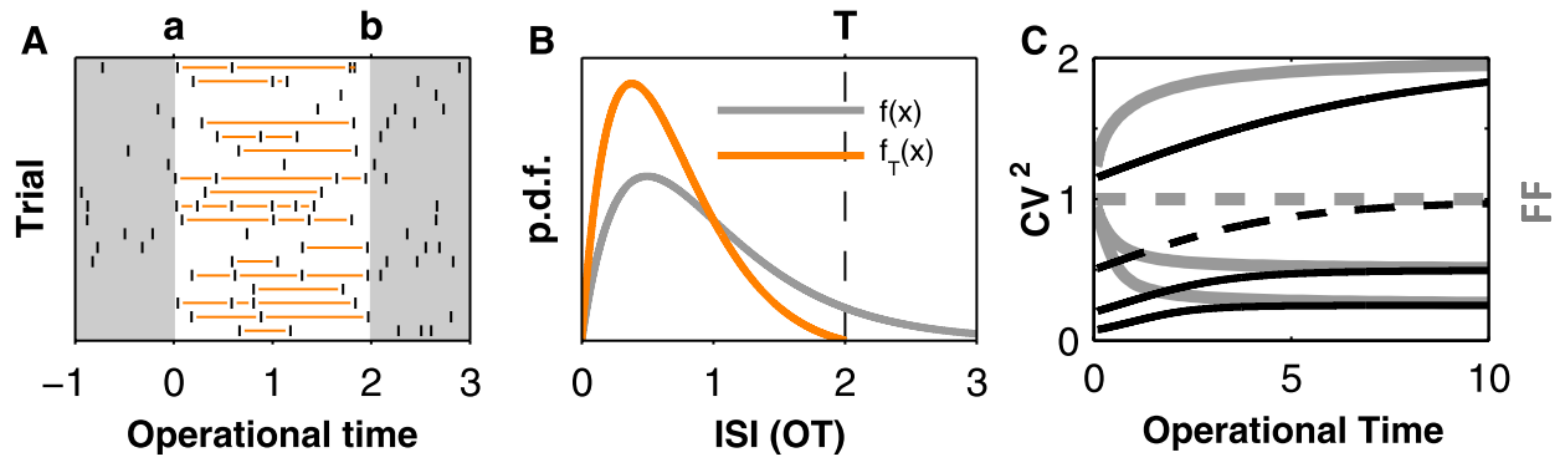
where ξ_i denote coefficients of serial interval correlation.

Thus, for the equilibrium **renewal processes** we expect:

$$FF \approx CV^2$$

Empirical Estimation: Bias for finite observation T

- **Bias of empiric CV and FF** (as a function of operational time)
- Practical consequence: use **large windows ≥ 5 ISIs**



$$\hat{f}(x) = \begin{cases} \eta^{-1}(T - x)f(x) & \text{for } x \in [0, T], \\ 0 & \text{otherwise,} \end{cases}$$

$$\eta = \int_0^T (T - s)f(s) ds$$

Nawrot et al. (2008) *J Neurosci Meth* 169

Nawrot (2010) In: Grün, Rotter (eds.), *Springer Series Comp Neurosci* 7

Take Home's - I

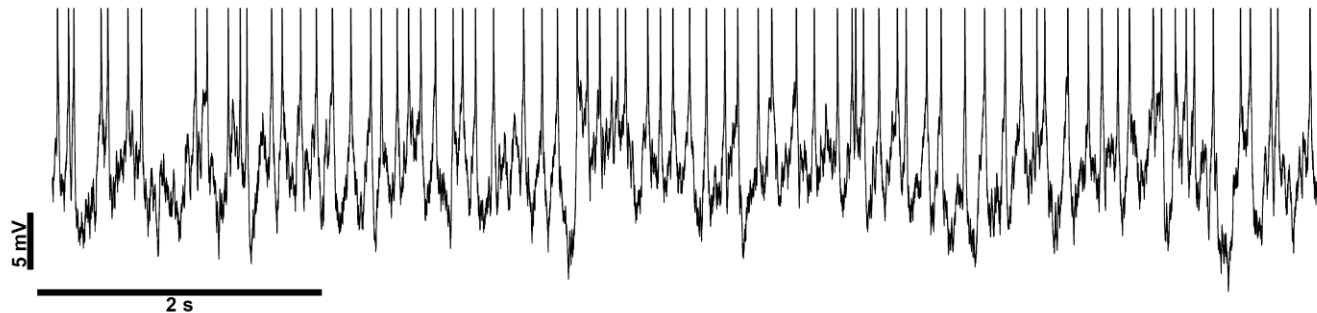
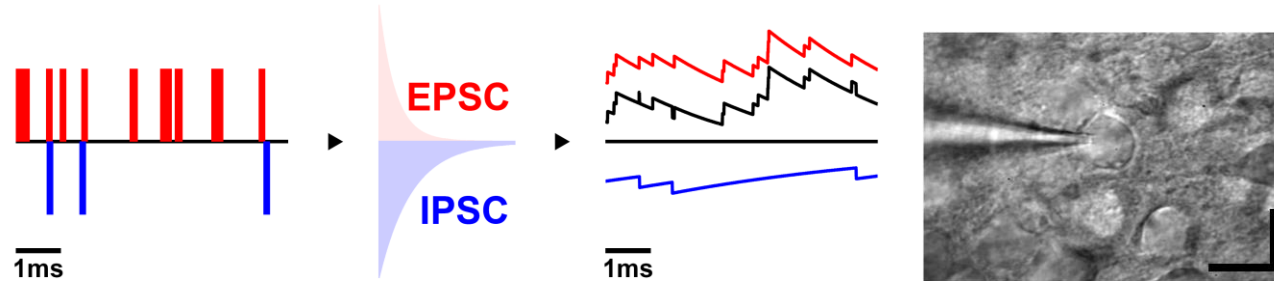
- Point process theory links *interval* and *counting* statistics
renewal process : $FF \approx CV^2$ (Poisson process : $FF = CV^2 = 1$)
- Fano factor and CV exhibit a bias for short observation times T
 $\lim_{T \rightarrow 0} (FF) = 1$, $\lim_{T \rightarrow 0} (CV^2) = 0$

Part I

Empirical Analysis of Variability in Neuronal Spike Trains

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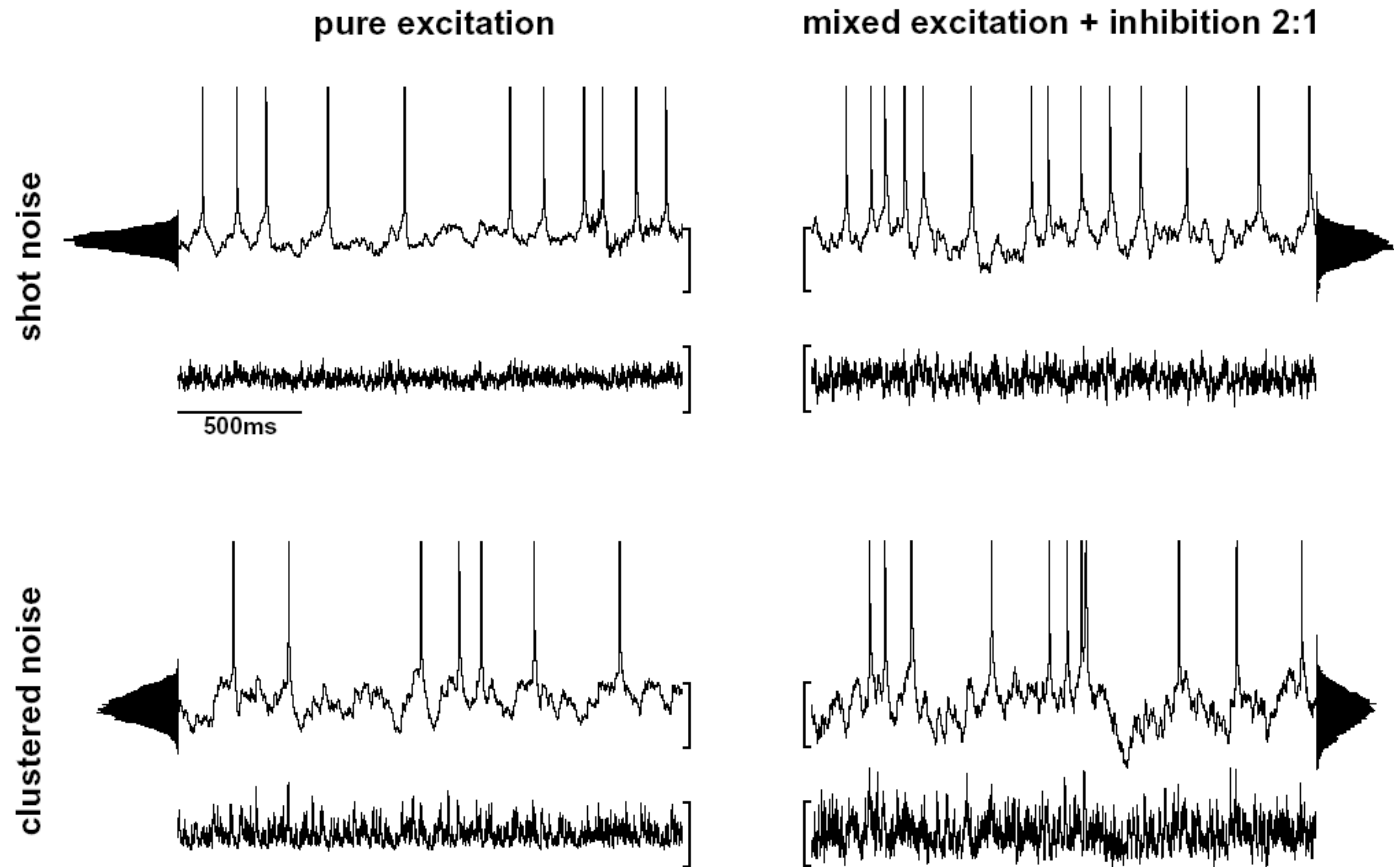
In Vitro Noise Current Injection (stationary)



Experiments with **Clemens Boucsein** & **Victor Rodriguez**

University of Freiburg, Germany

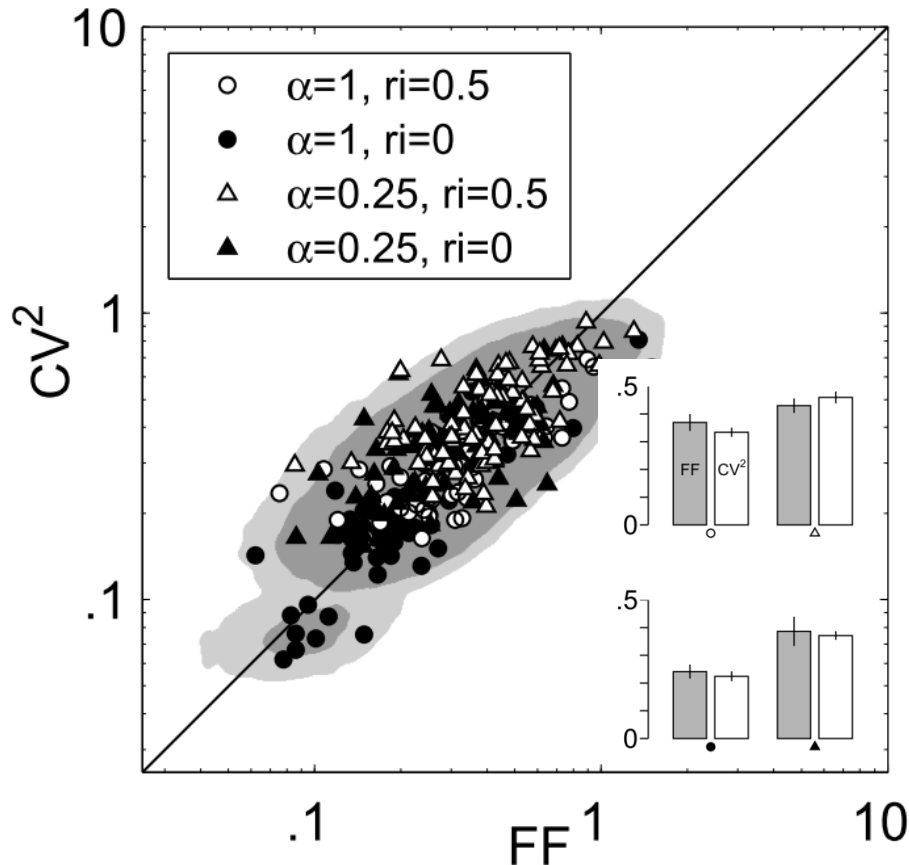
In Vitro Noise Current Injection



Nawrot et al. (2007) Neurocomputing 70
Nawrot et al. (2008) J Neurosci Meth 169

In Vitro Noise Current Injection

- ▶ Cortical neurons are clearly **less variable than the Poisson process**
- ▶ Renewal-like behavior with **$FF \approx CV^2$**



$$FF \approx CV^2 \leq 0.5$$

CV - Holt et al. (1996) *J Neurophysiol* 75; Nowak et al. (1997) *Cereb Cortex* 7; Stevens & Zador (1998) *Nat Neurosci* 1; Harsch & Robinson (2000) *J Neurosci* 20; Badoual et al. (2005) *Neurocomp* 65; Arsiero et al. (2007) *J Neurosci* 27

FF - Stevens & Zador (1998) *Nat Neurosci* 1; Harsch & Robinson (2000) *J Neurosci* 20

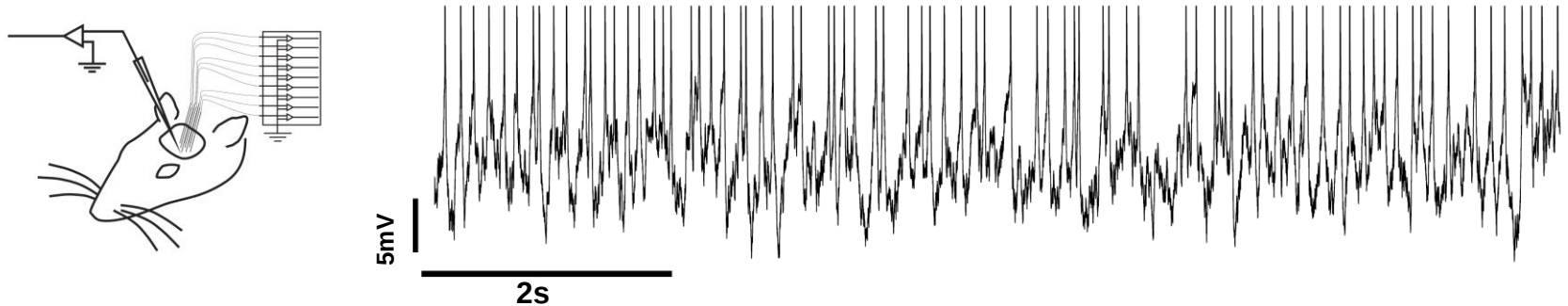
Nawrot (2003) PhD Thesis

Nawrot et al. (2008) *J Neurosci Meth* 169

Nawrot (2010) *Analysis of Parallel Spike Trains*

In Vivo Spontaneous Activity (stationary)

- *In vivo* **intracellular** recordings, somatosensory cortex of anesthetized rats
- **spontaneous activity** (no stimulation)

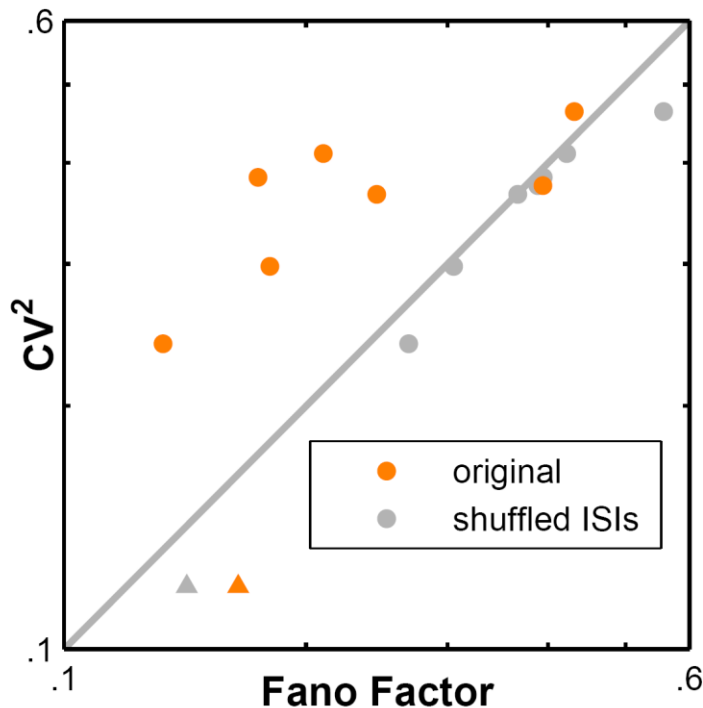


Experiments by **Clemens Boucsein & Dymphie Suchanek**

University of Freiburg, Germany

In Vivo Spontaneous Activity

- ▶ Cortical neurons *in vivo* show **small FF ≈ 0.25**
- ▶ Non-renewal behavior **FF $< CV^2$** (a *sink* of variability)



$$FF < CV^2 \leq 0.5$$

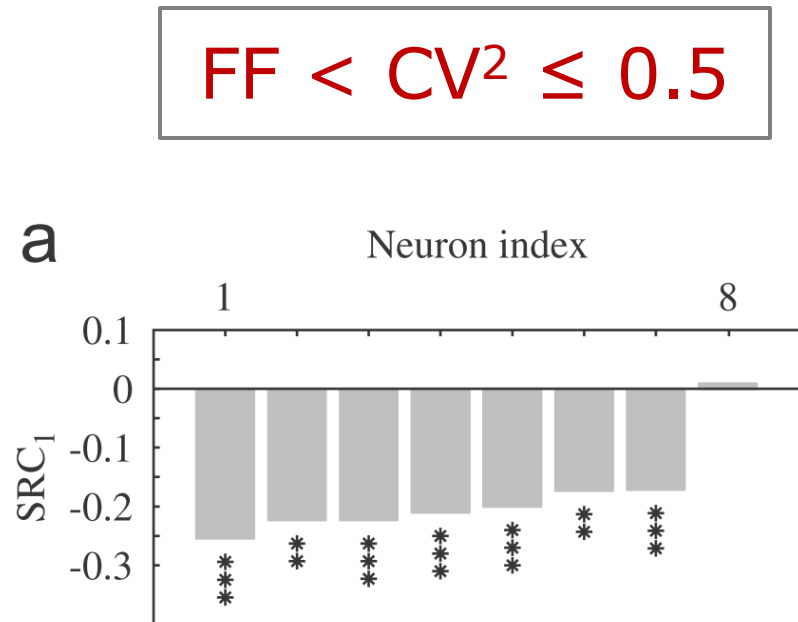
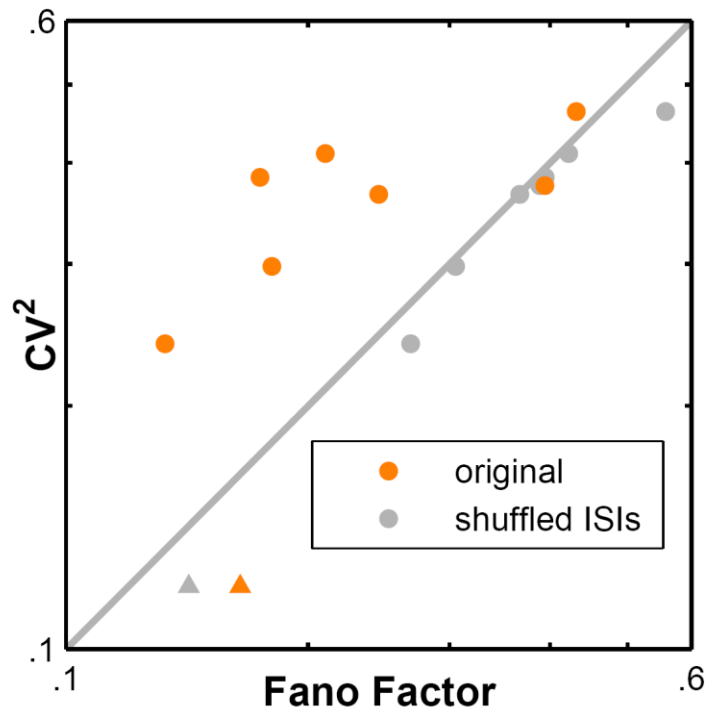
$$\lim_{T \rightarrow \infty} FF = CV_{\infty}^2 (1 + 2\xi)$$

Nawrot et al. (2007) *Neurocomputing* 70

Farkhooi, Strube-Bloss, Nawrot (2009) *Phys Rev E* 79

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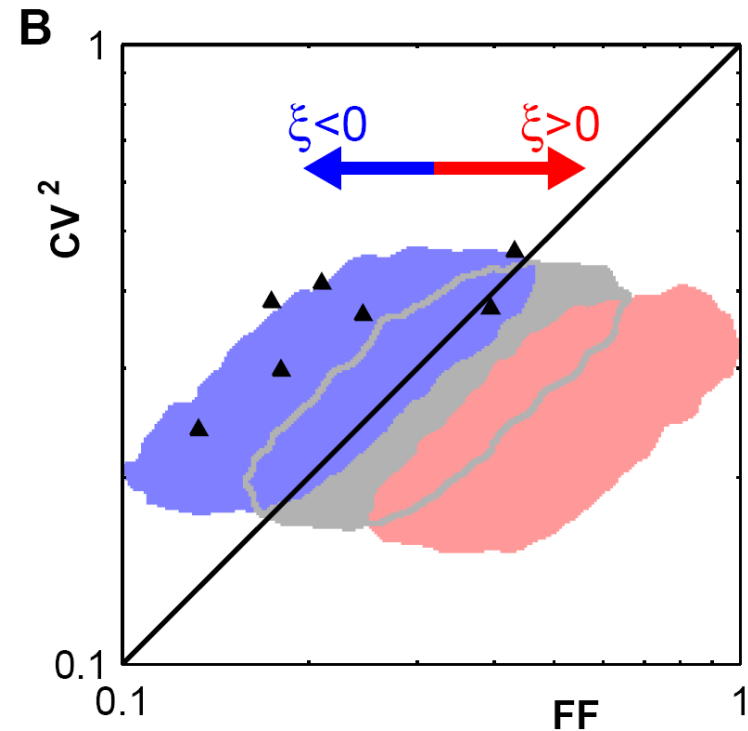
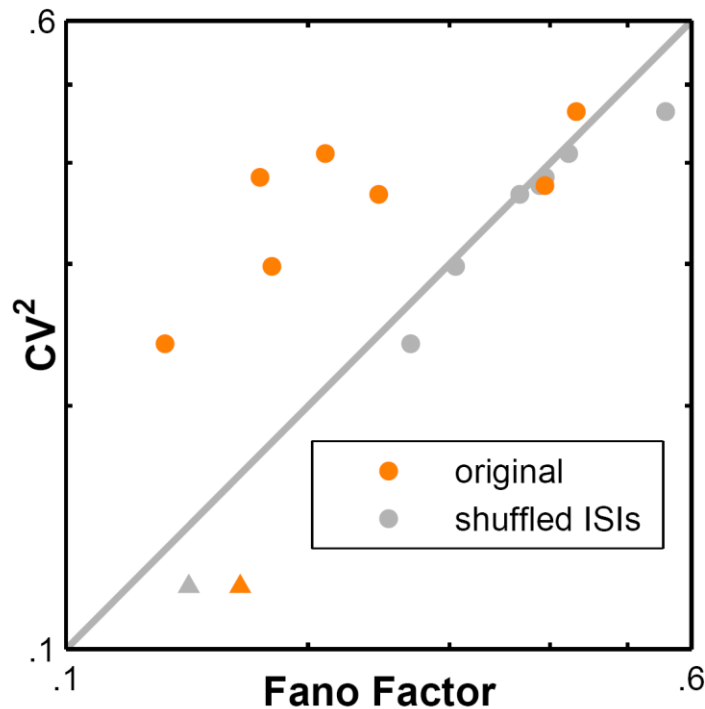


Nawrot et al. (2007) Neurocomputing 70

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In Vivo Spontaneous Activity

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Nawrot et al. (2007) Neurocomputing 70
Nawrot (2010) Analysis of Parallel Spike Trains

Take Home's - II

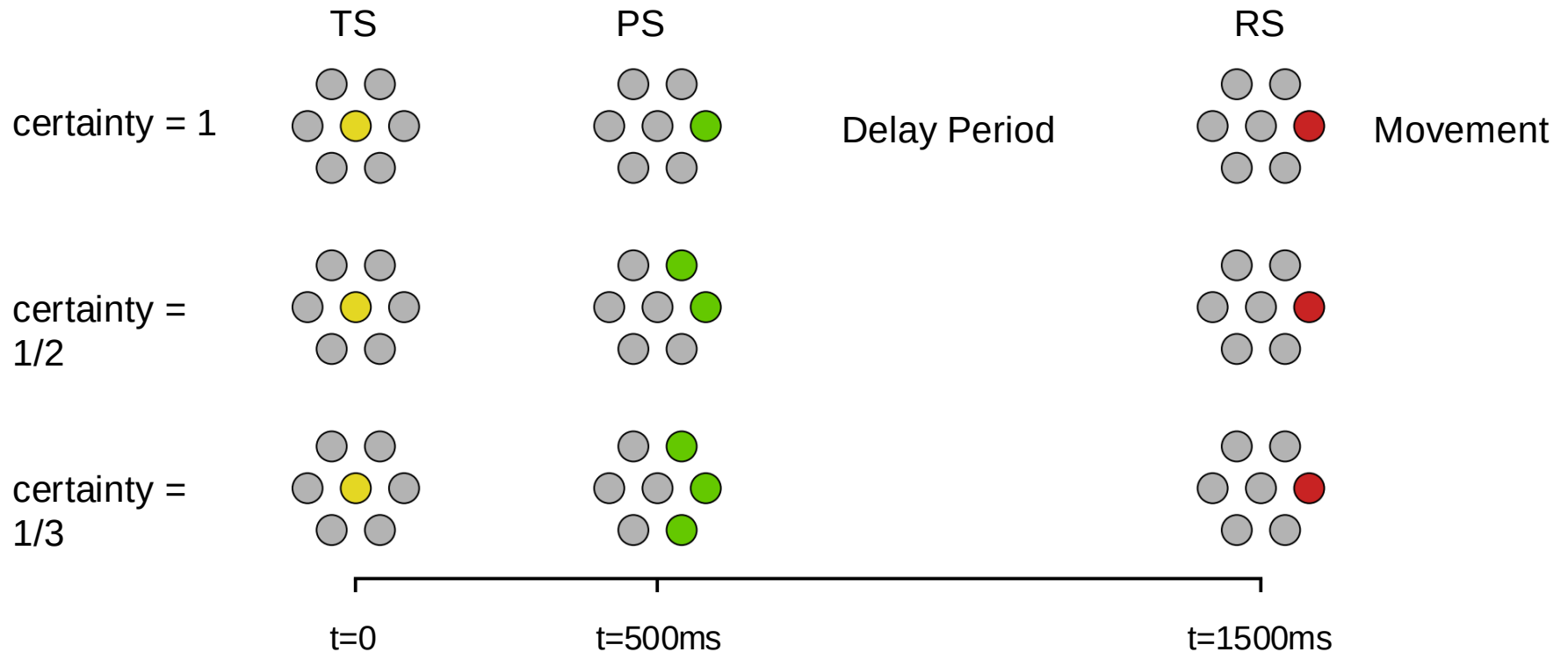
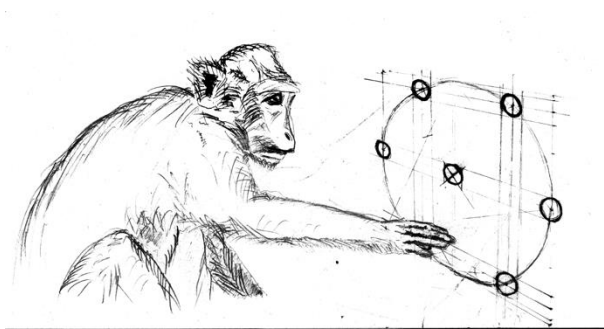
- Under **stationary input conditions** (balanced network) cortical neurons are *less* variable than the Poisson process with $FF < CV^2 \approx 0.5$
- **Negative** serial interval correlations **reduce count variability** at the single neuron and the population level
- **Poisson process not a good model** for single cortical neurons, which exhibit lower output variability, refractoriness, and non-renewal properties ($CV \sim 0.5 \mid \xi < 0$)

Part I

Empirical Analysis of Variability in Neuronal Spike Trains

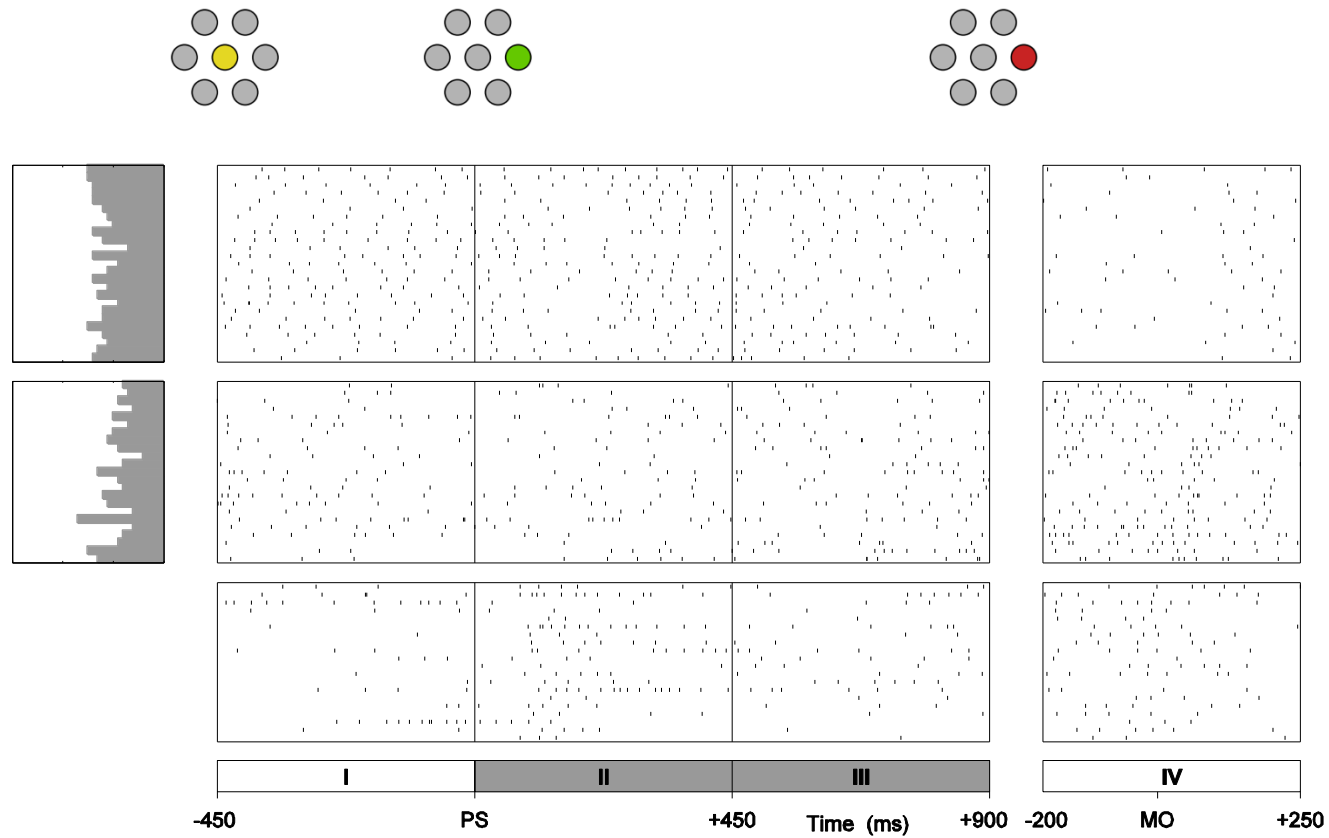
- I Measuring Interval and Count Variability
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Delayed Reach-Task with Varying Uncertainties



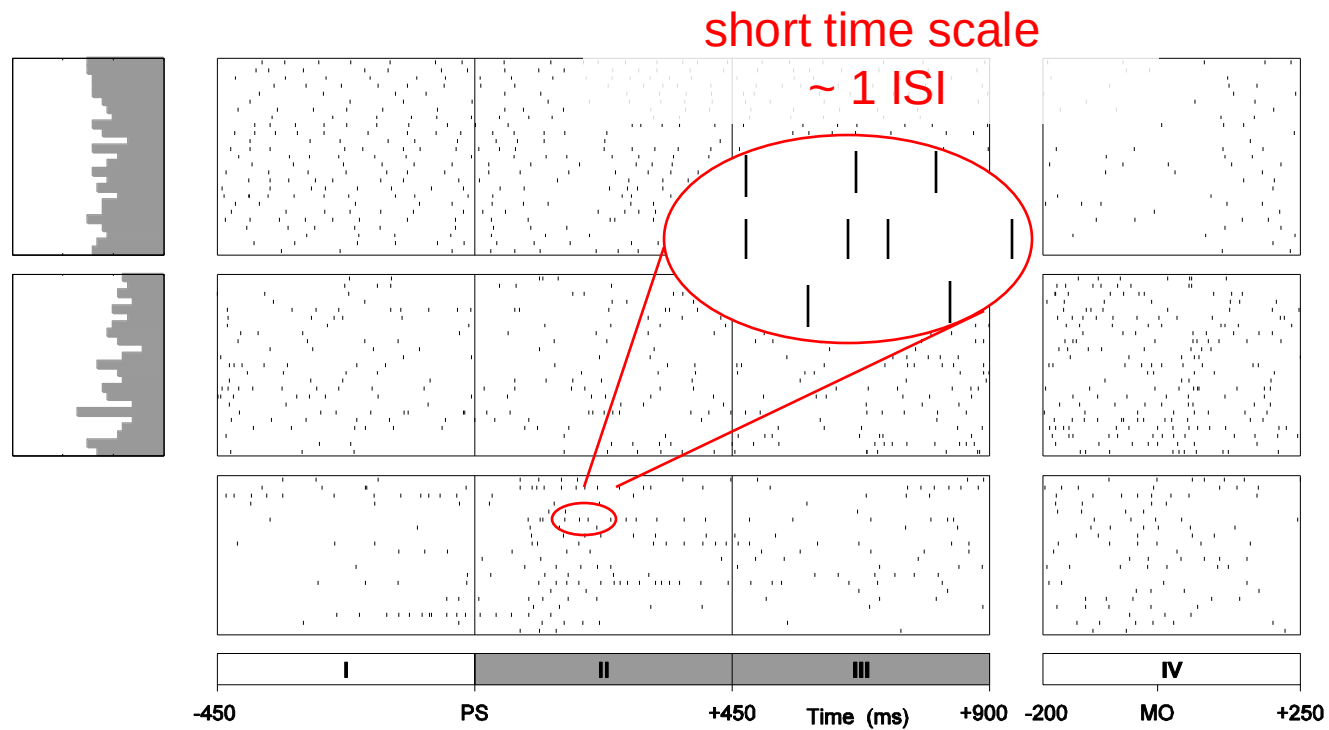
Experiments by **Alexa Riehle**, CNRS & Université Marseille (France) & Jülich Research Center (Germany)

Task-related Variability in the Motor Cortex



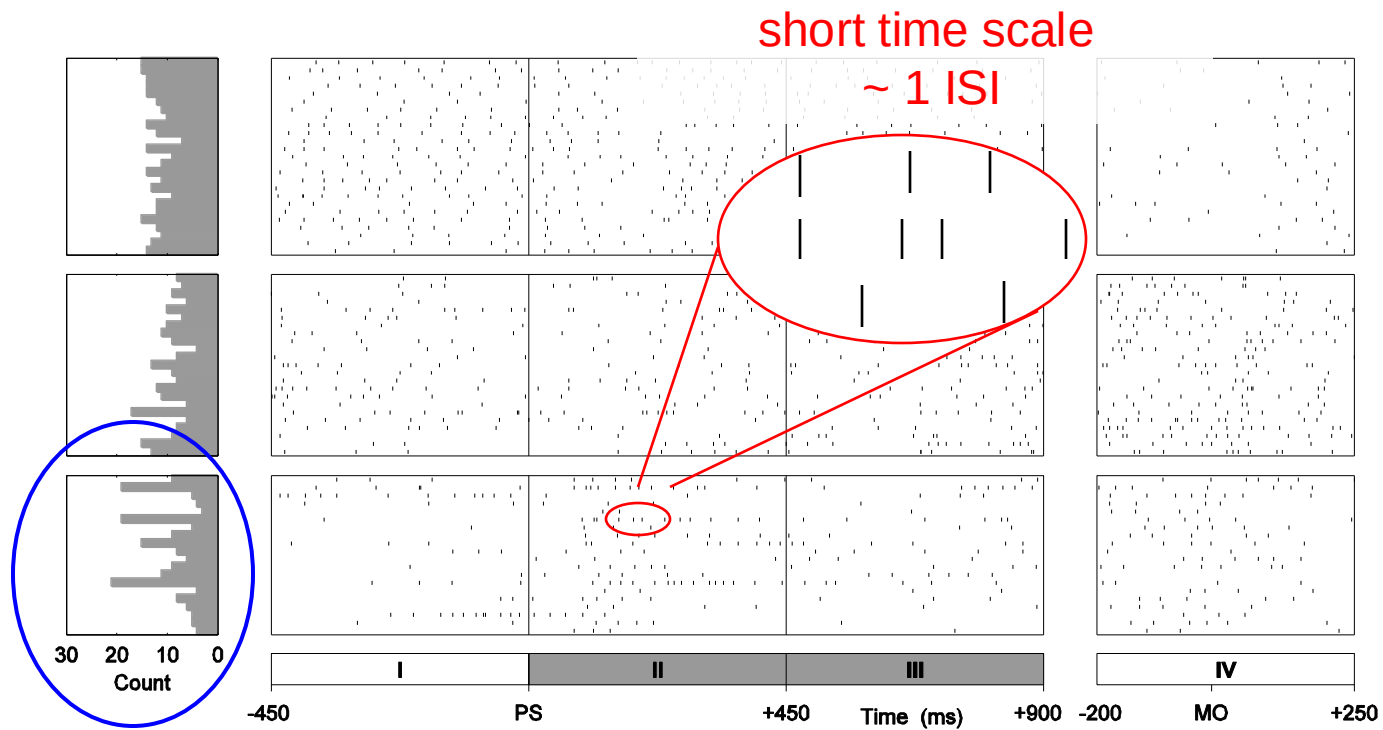
Task-related Variability in the Motor Cortex

► Interval variability: CV2



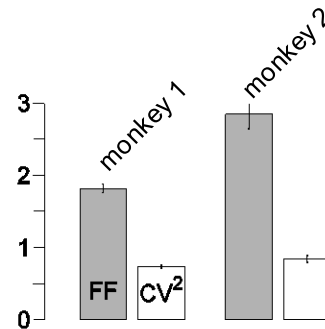
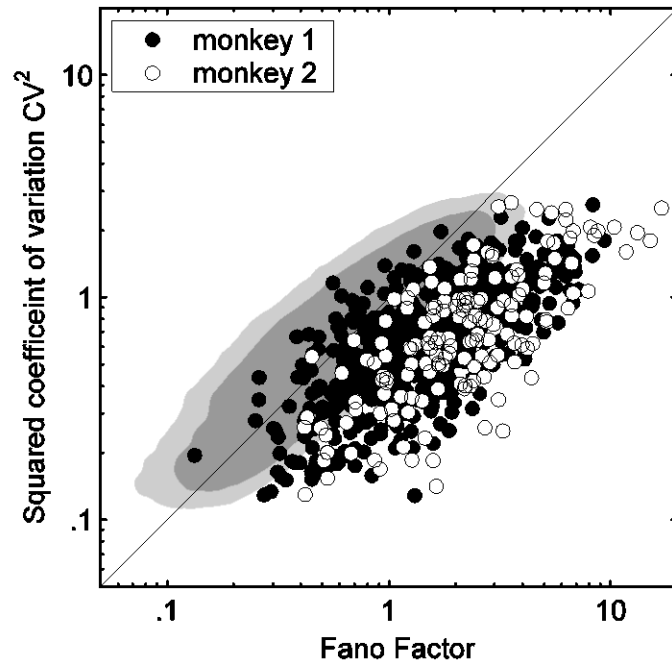
Task-related Variability in the Motor Cortex

- ▶ Interval variability: CV^2
- ▶ Spike count variability: FF

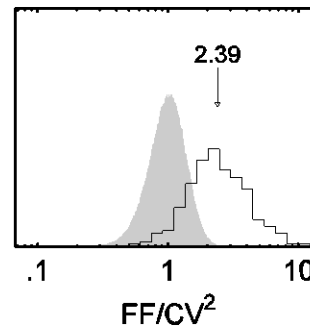


Task-related Variability in the Motor Cortex

- cortical neurons *in vivo* disagree with renewal prediction

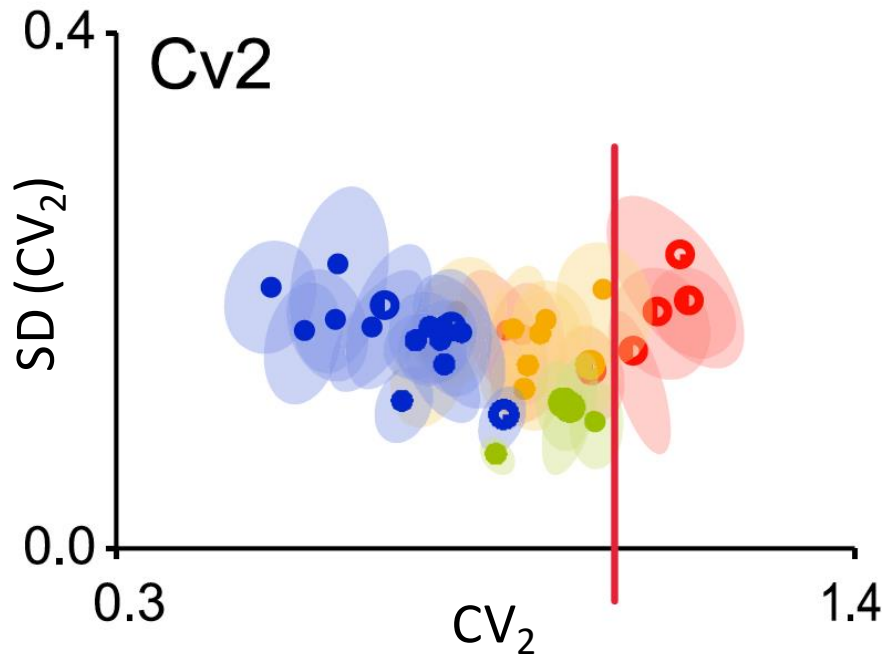
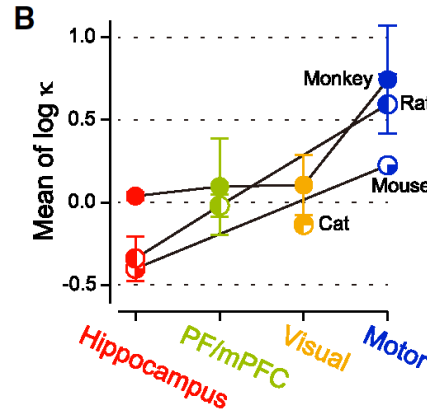
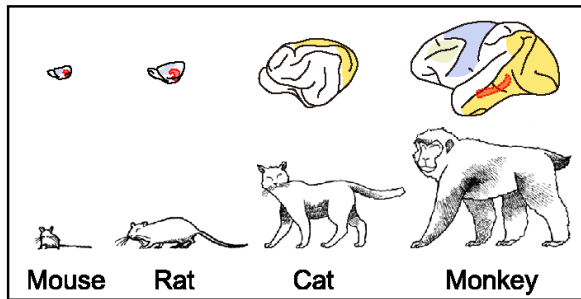


$$FF \gg CV^2$$



Nawrot (2010) In: Grün, Rotter (eds.), Springer Series Comp Neurosci 7
Riehle, Brochier, Nawrot, Grün (2018) Front Neural Circuits 12

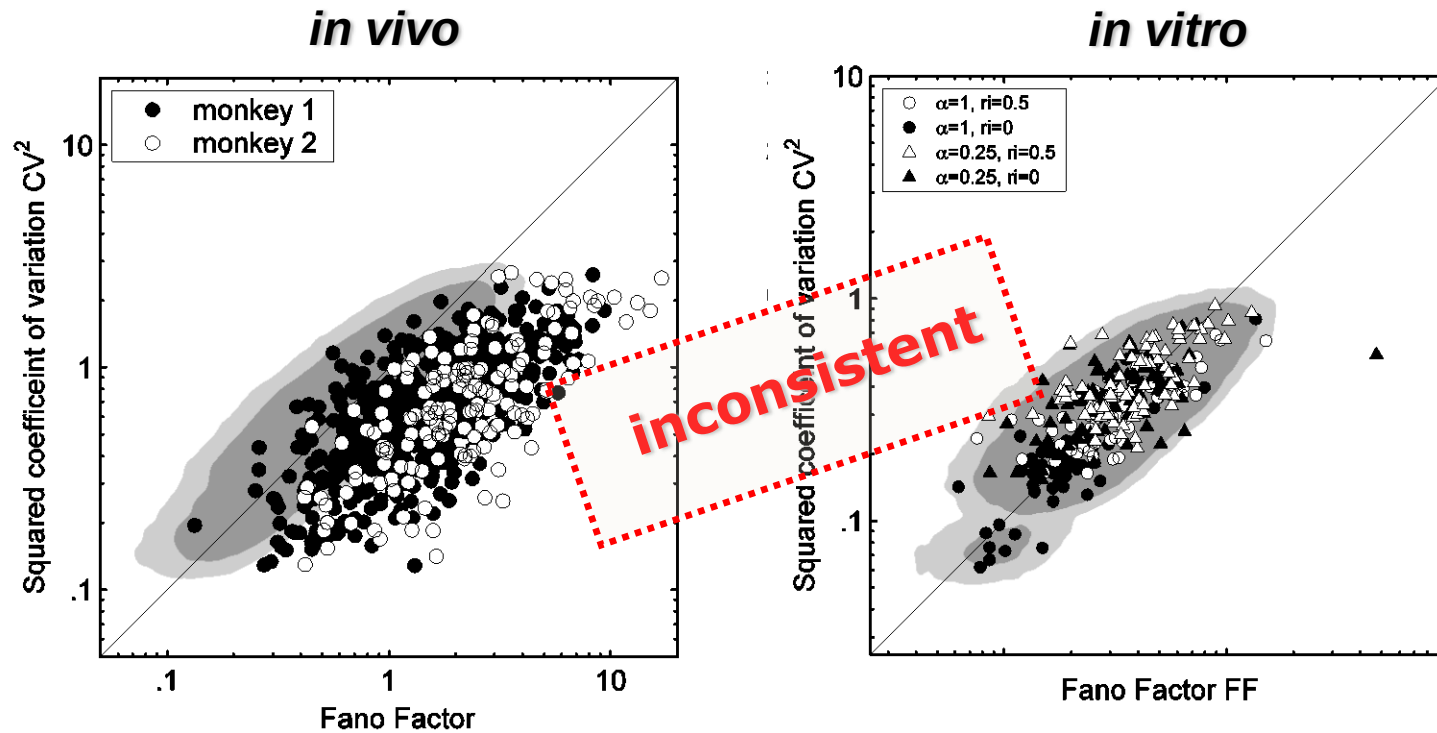
Interval variability is higher in sensory than in motor areas



○ : Mouse ◐ : Rat ◑ : Cat ● : Monkey

Task-related Variability in the Motor Cortex

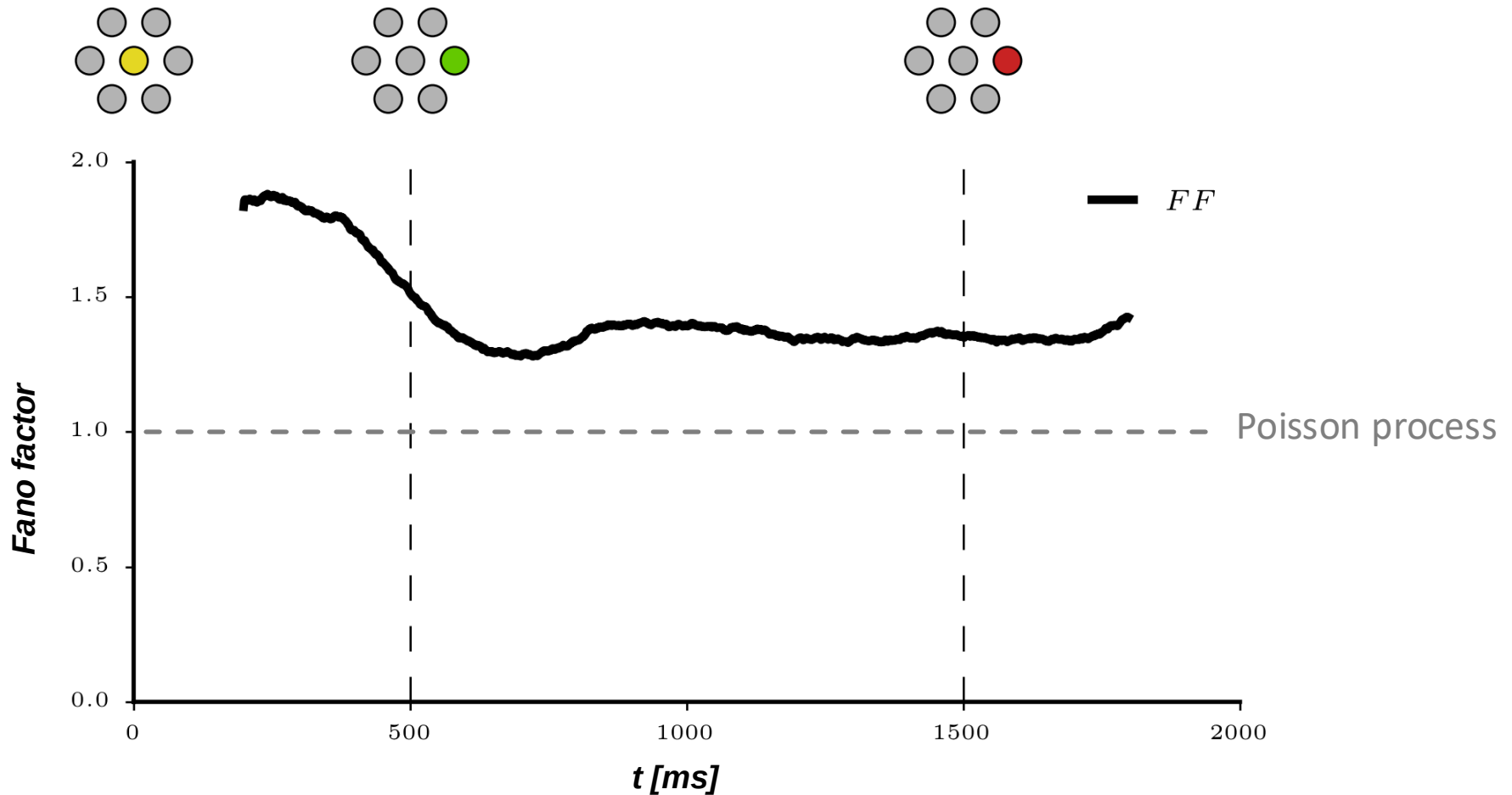
- ▶ cortical neurons *in vivo* disagree with renewal prediction
- ▶ in contradiction to the stationary input situation



Nawrot (2010) In: Grün, Rotter (eds.), Springer Series Comp Neurosci 7

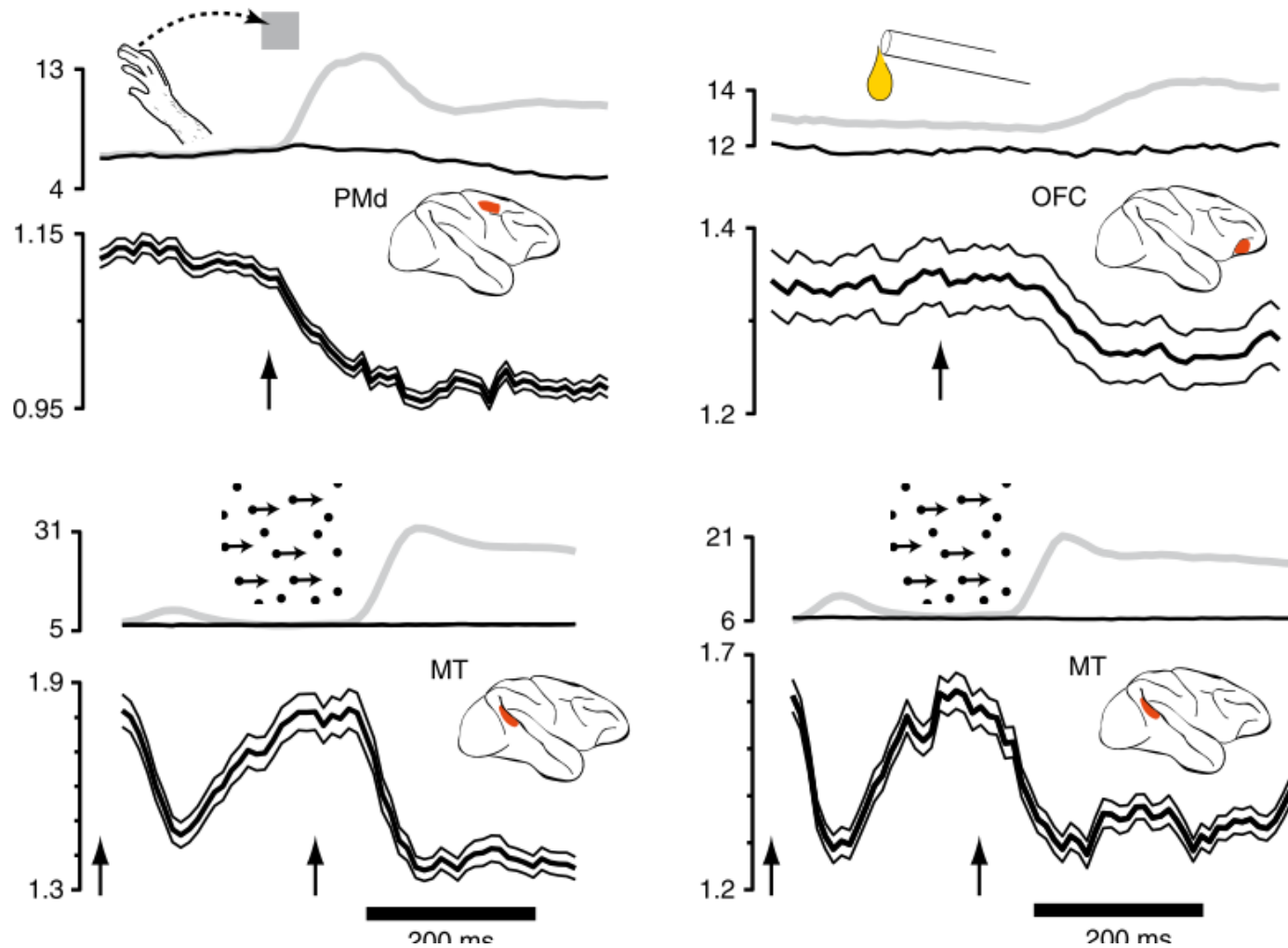
Nawrot et al. (2008) J Neurosci Meth 169

Task-related Variability Dynamics *in vivo*



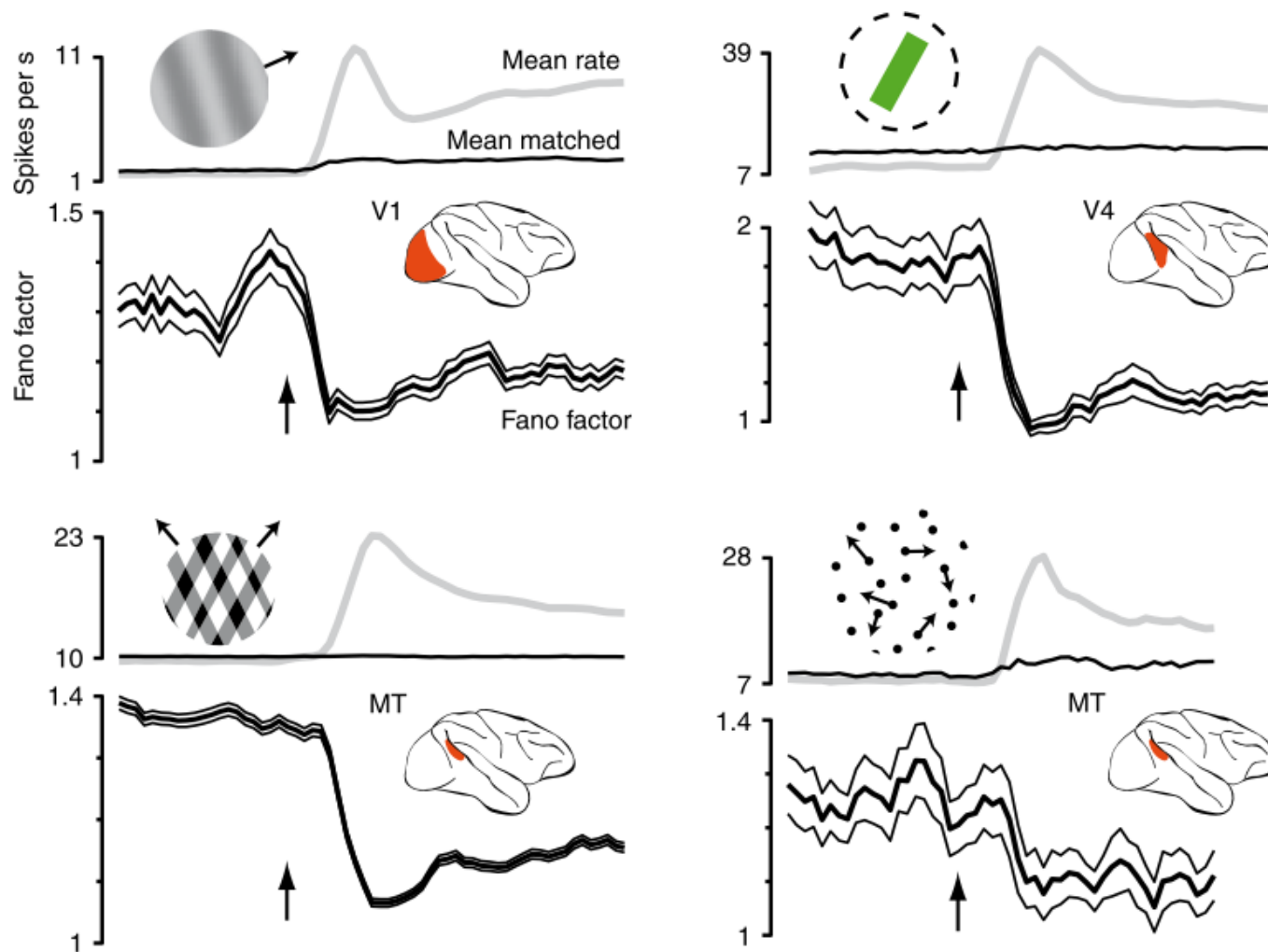
Rostami et al. (2024) Nat Commun
Rickert et al. (2009) J Neurosci
Nawrot (2003) PhD Thesis

Task-related Variability Dynamics *in vivo*



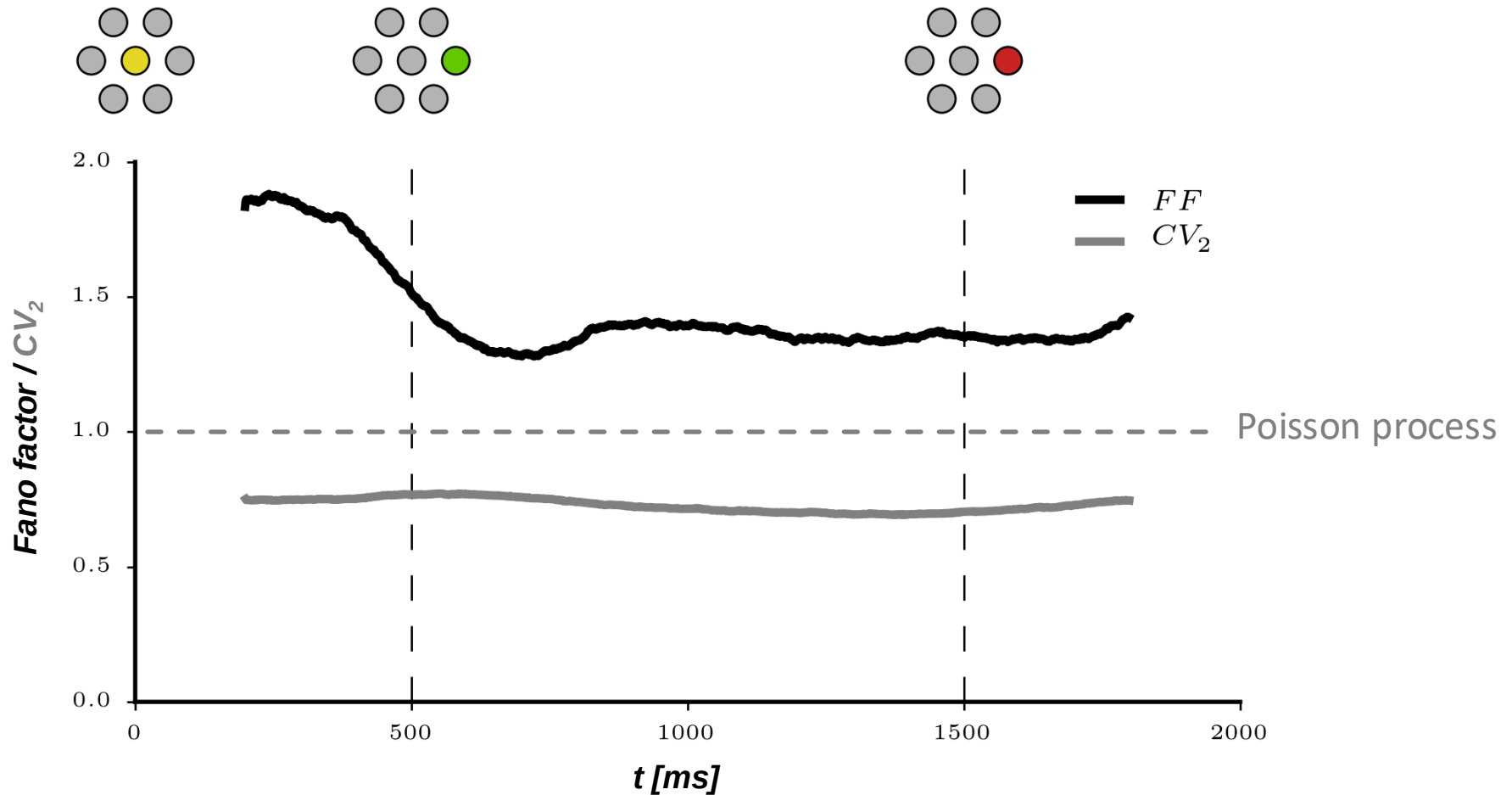
Churchland et al. (2010) Nat Neurosci 13
Churchland et al. (2006) J Neurosci

Task-related Variability Dynamics *in vivo*



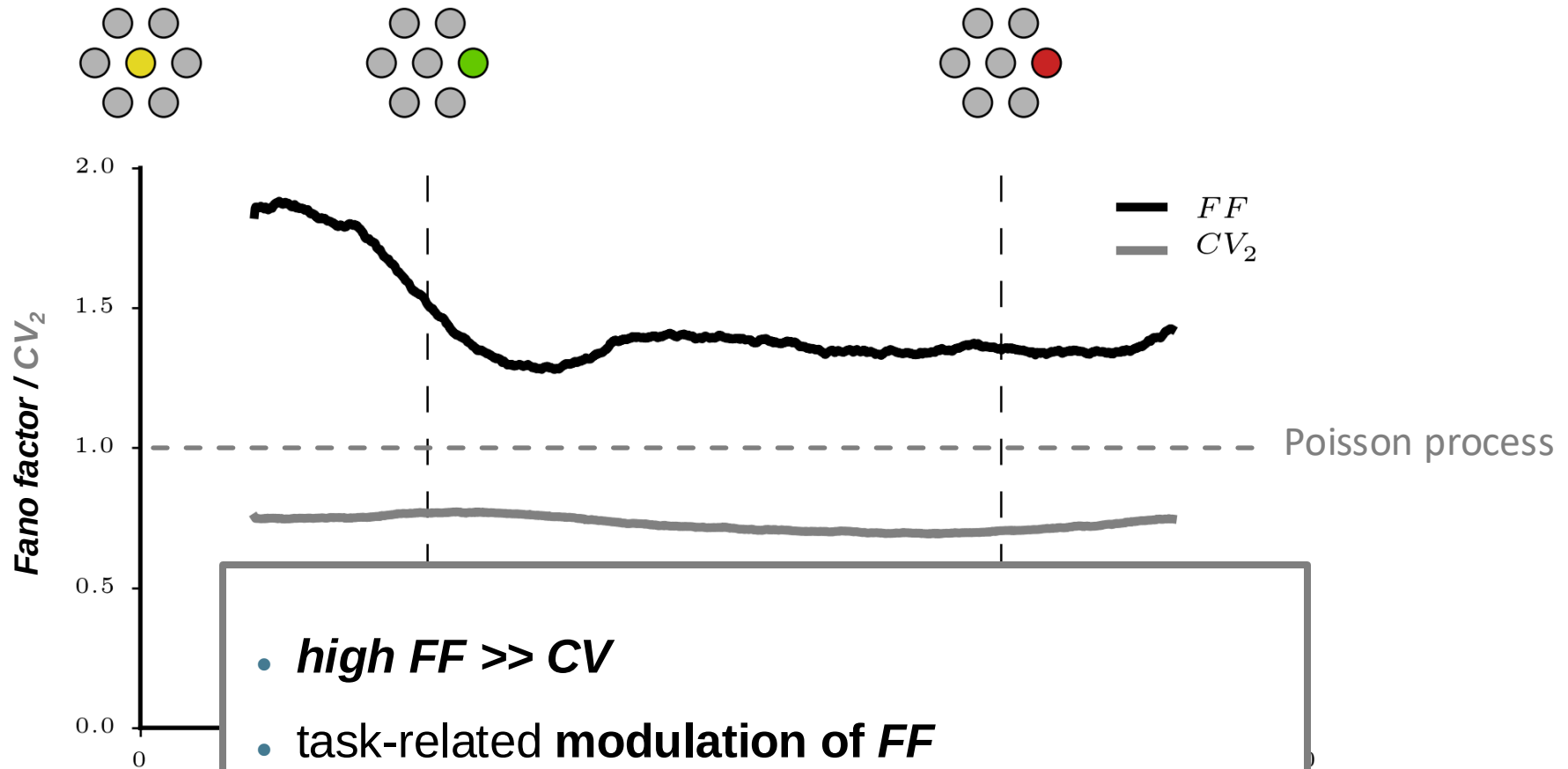
Churchland et al. (2010) Nat Neurosci 13
Churchland et al. (2006) J Neurosci

Variability Dynamics *in vivo*



Rostami et al. (2024) Nat Commun
Riehle et al. (2018) Front Neural Circuits

Variability Dynamics *in vivo*



- ***high $FF \gg CV$***
 - **task-related modulation of FF**
 - **CV is constant**
- ➔ Which mechanisms rule this dynamics ?**

Take Home's - III

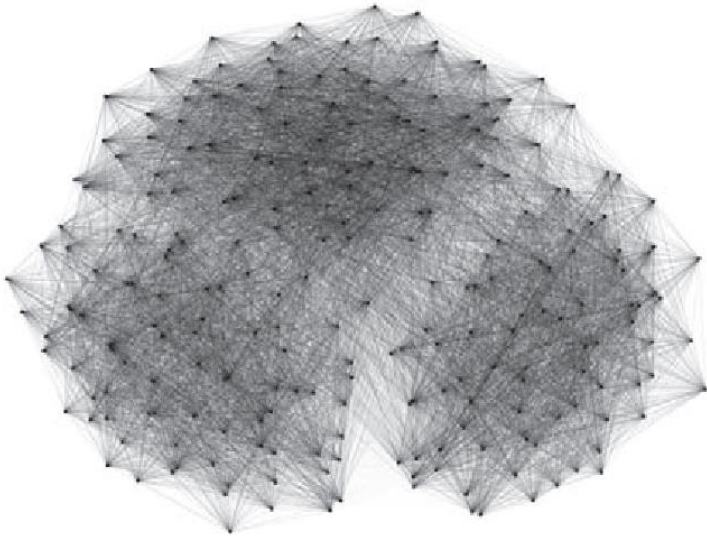
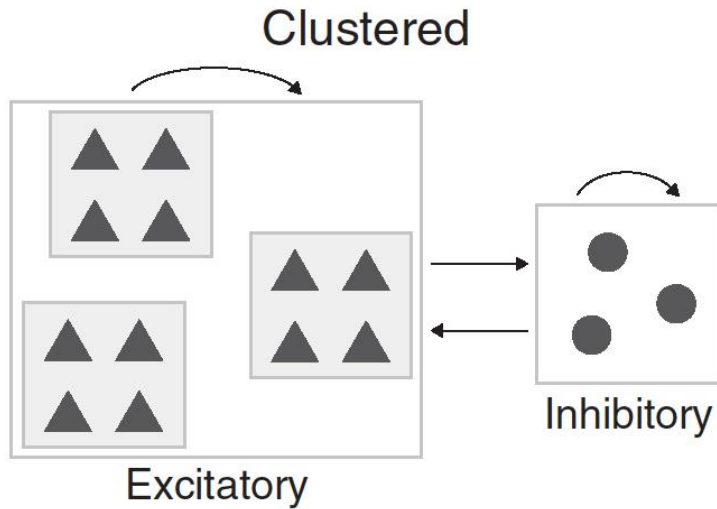
- Trial-by-Trial Variability in Motor Cortex is Large $FF > 1.5$
(and typically larger than in sensory areas)
- Count Variability is larger than Interval Variability $FF > CV^2$
- Time-resolved Fano factor reduces during task
- Time-resolved Coefficient of Variation is constant throughout task

Part II

Spiking Attractor Model Explains Cortical Variability Dynamics

**What are the mechanisms to explain
high trial-to-trial variability versus
low interval variability in the brain?**

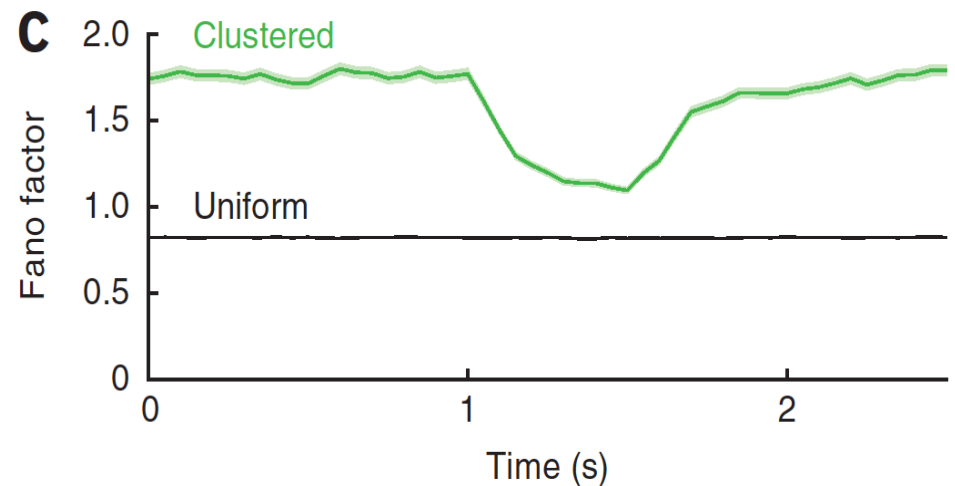
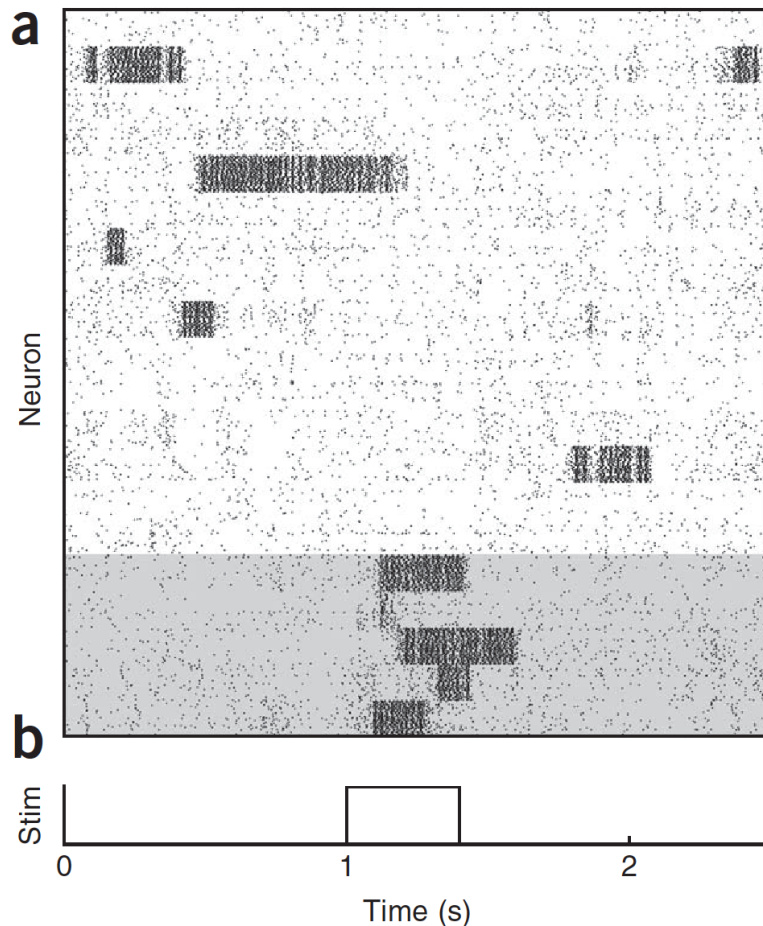
Attractor Dynamics in Clustered Excitation Model



Deco et al (2012) PLoS CB
Litwin - Kumar & Doiron (2012) Nat. Neurosci
Mazzucato et al (2015) J Neurosci
Mazzucato et al (2019) Nat Neurosci

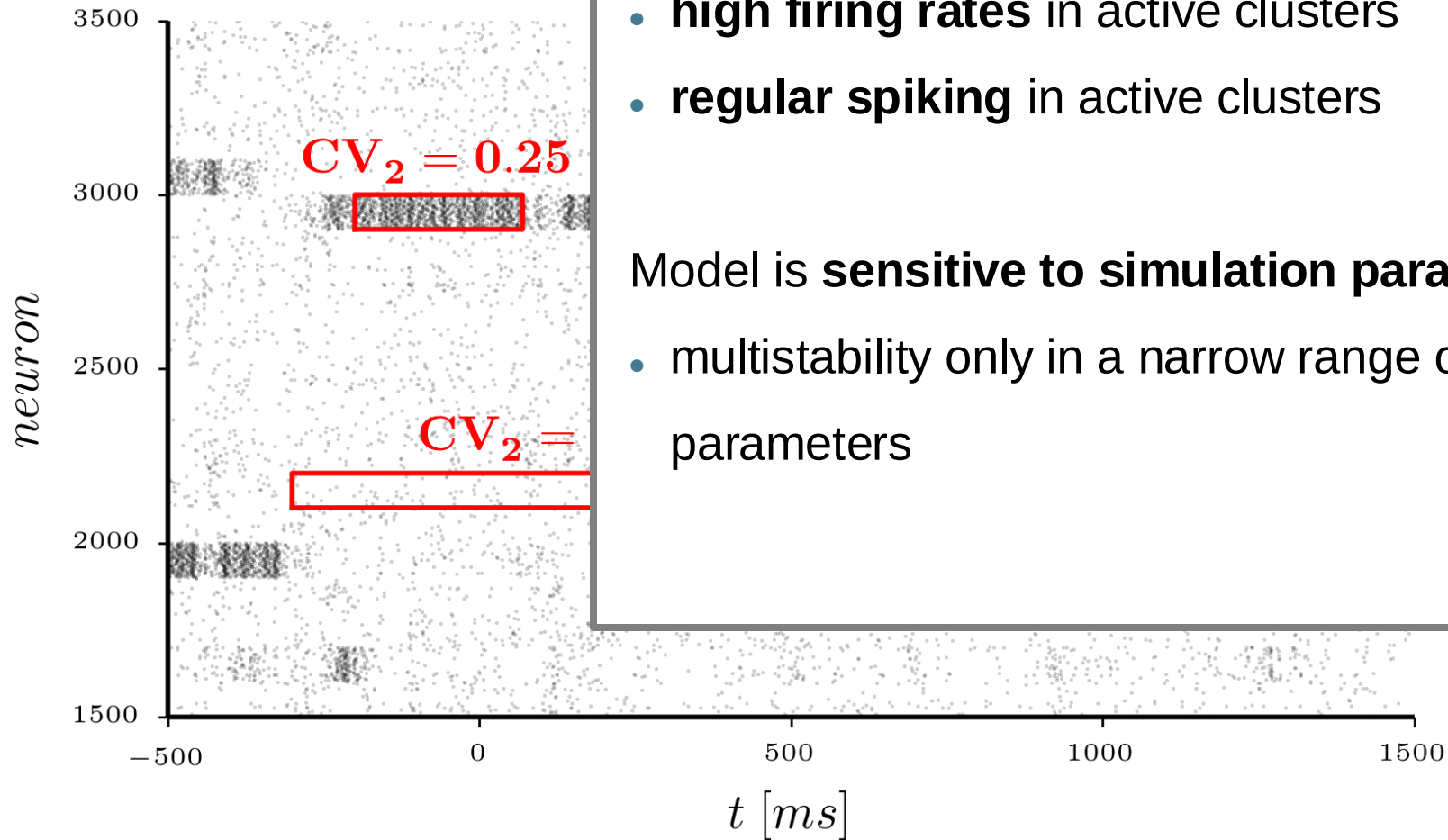
Attractor Dynamics in Clustered Excitation Model

- ▶ in spont. condition the clustered network exhibits **attractor dynamics**
- ▶ cluster cycling generates **across trial non-stationarity** (high FF)
- ▶ stimulation results **in temporary suppression of FF**



Litwin - Kumar & Doiron (2012) Nat. Neurosci

Attractor Dynamics in Clustered Excitation Model



Remaining Inconsistencies with exp. data:

- **high firing rates** in active clusters
- **regular spiking** in active clusters

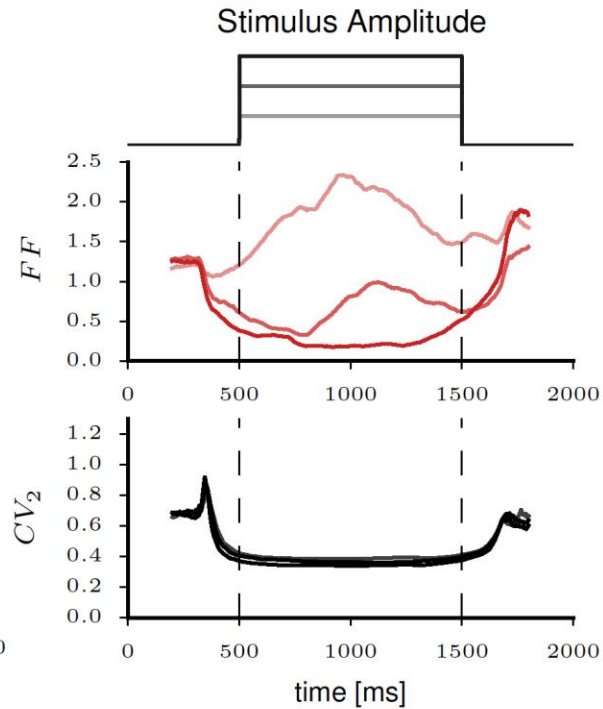
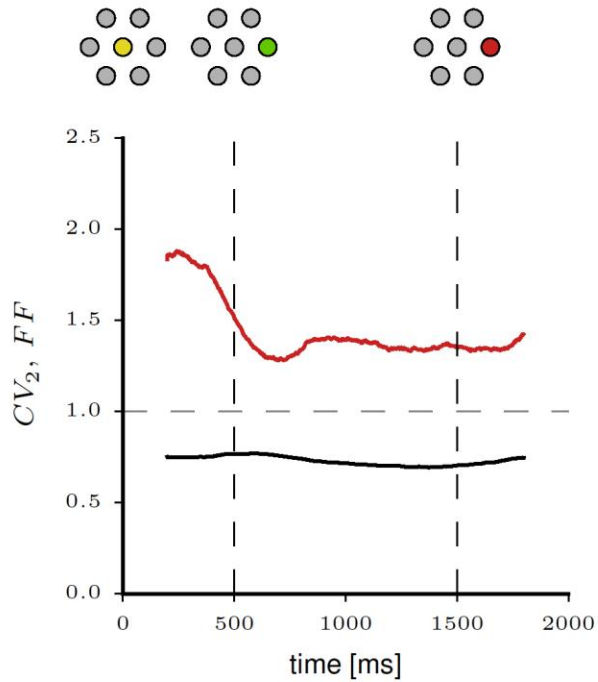
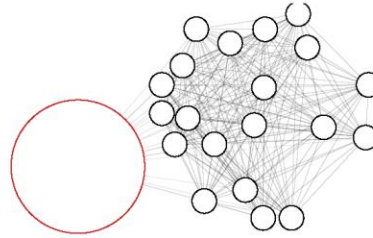
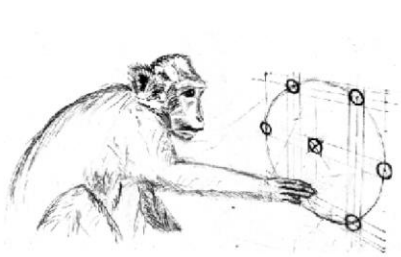
Model is **sensitive to simulation parameters**

- multistability only in a narrow range of parameters

Litwin - Kumar & Doiron (2012) Nat. Neurosci

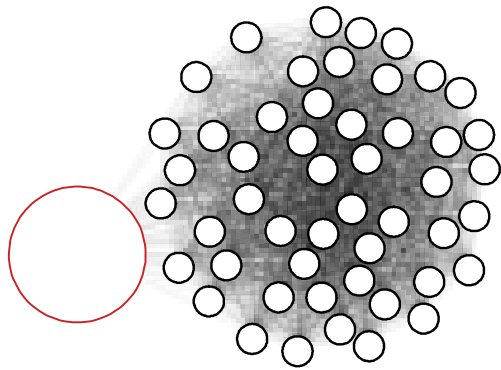
Attractor Dynamics in Clustered Excitation Model

E - clustering



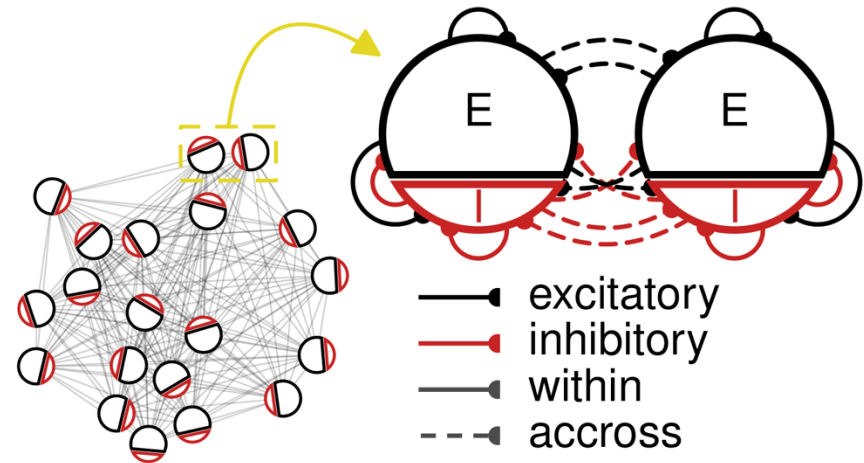
Excitatory-Inhibitory (E/I) Cluster Topology

E - clustering



Song et al (2005) PLoS Biol.
Yoshimura et al (2005) Nature
Perin et al. (2011) PNAS
Fino et al (2011) Neuron

E/I - clustering



Xue et al. (2014) Nature
Lee et al. (2014) Neuron
Trmblay et al. (2016) Neuron
Morishima et al. (2017) Cerebral Cortex
Znamenskiy (2018) BioRxiv
Khan et al (2018) Nat. Neurosci.
Najafi et al (2020) Neuron

Rostami et al. (2024) Nat Commun 15
Rost, Deger, Nawrot (2018) Biol Cyber 112

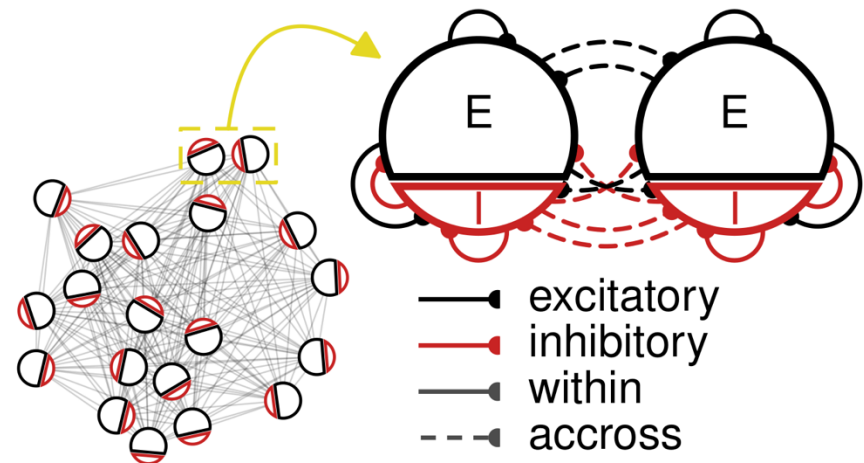
Excitatory-Inhibitory (E/I) Cluster Topology

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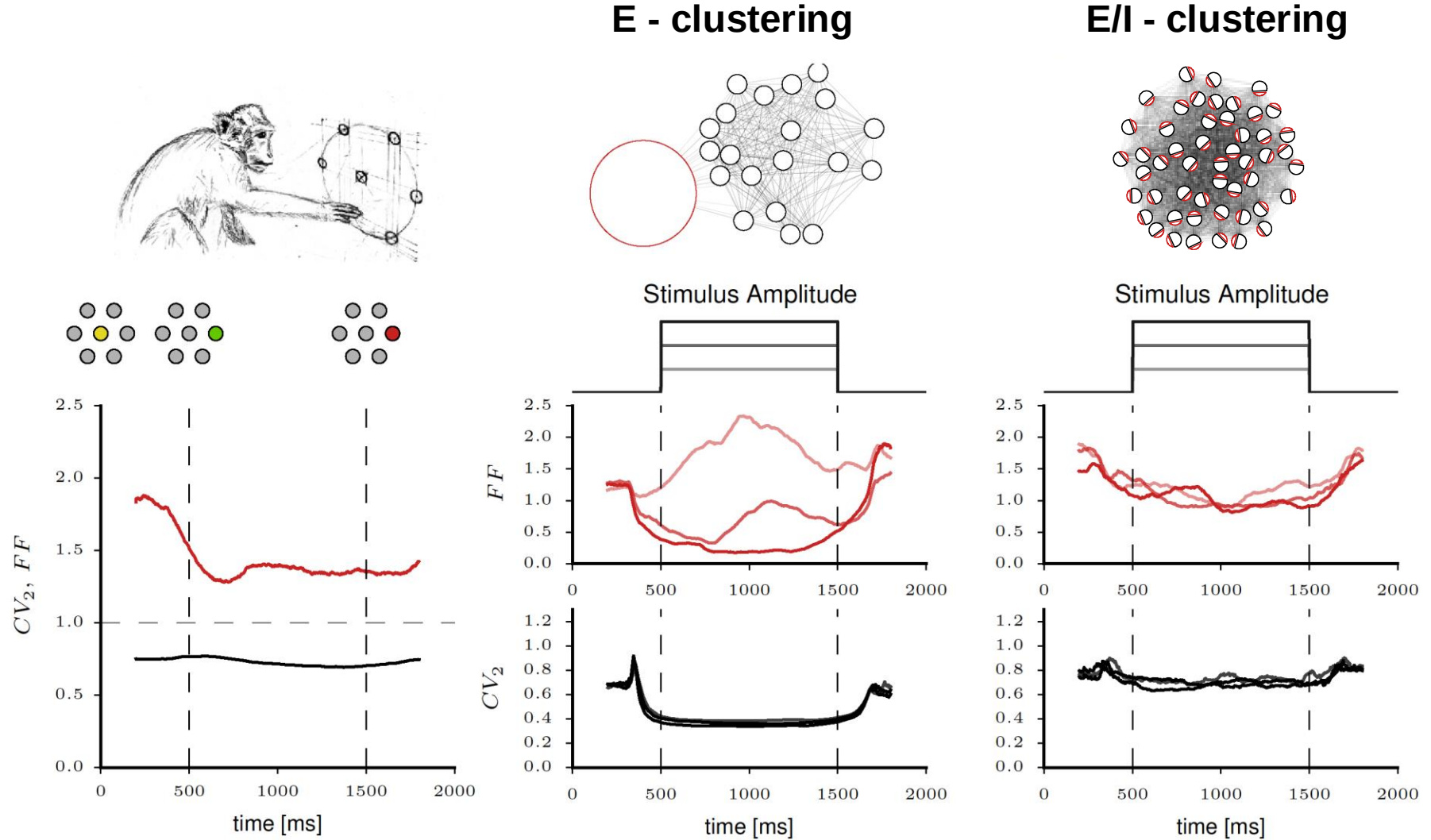
E/I - clustering

- Network divided into pairs of excitatory and inhibitory sub-populations
- Parameter for strength within/across clusters J_{E+}
- Parameter for ratio of excitatory and inhibitory cluster strength R_J
- Improves local E/I balance



Rostami et al. (2024) Nat Commun 15
Rost, Deger, Nawrot (2018) Biol Cyber 112

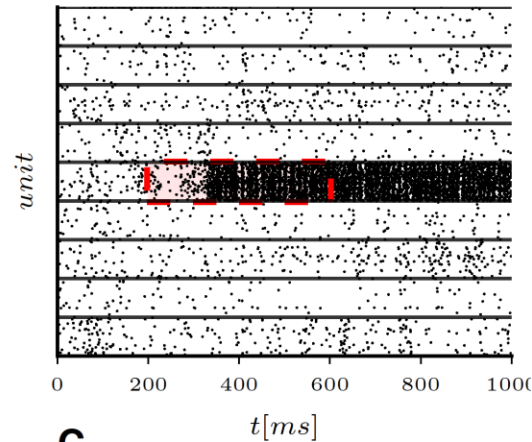
Reliable Attractor Activation and Constant CV in E/I Clustered Model



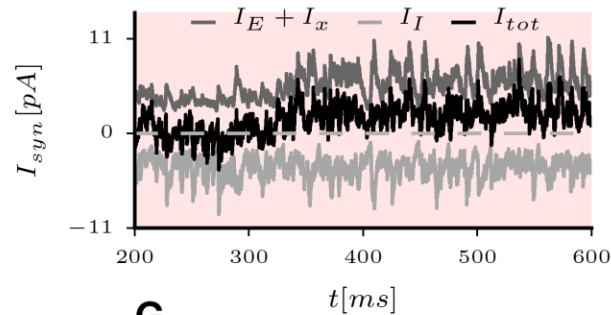
Local Balance Preserved in E/I Clustered Model

E - clustering

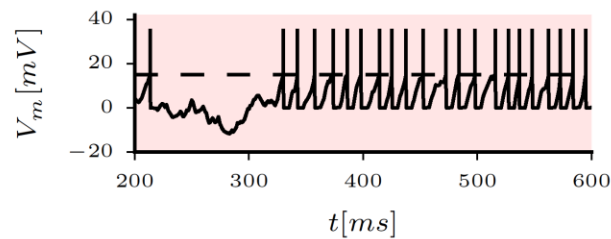
A



C



G



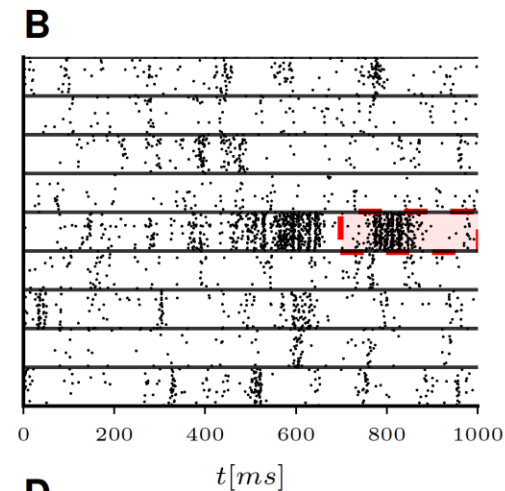
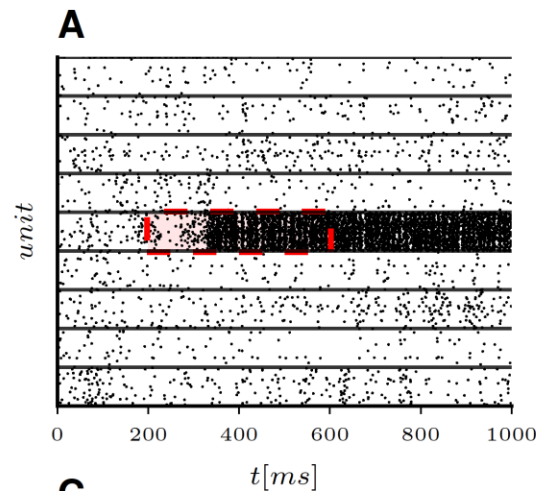
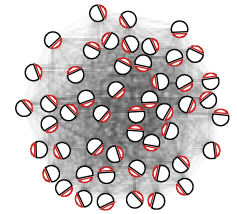
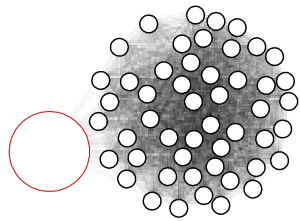
unbalanced

regular

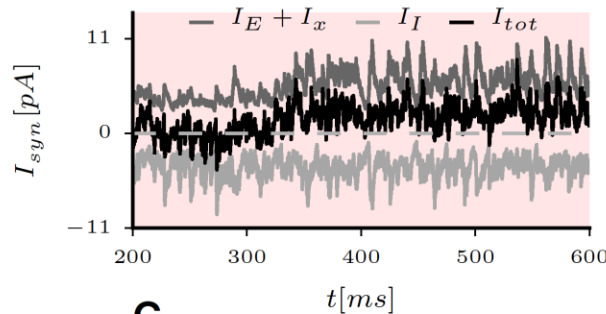
Local Balance Preserved in E/I Clustered Model

E - clustering

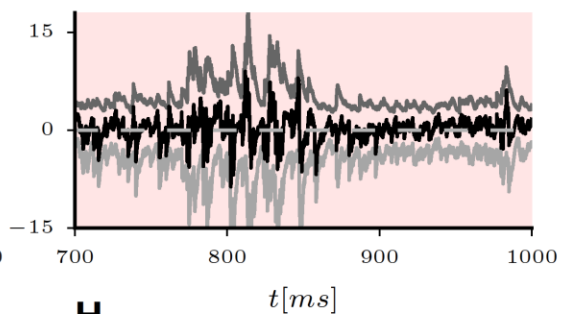
E/I - clustering



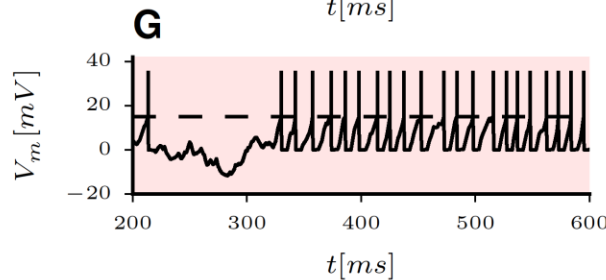
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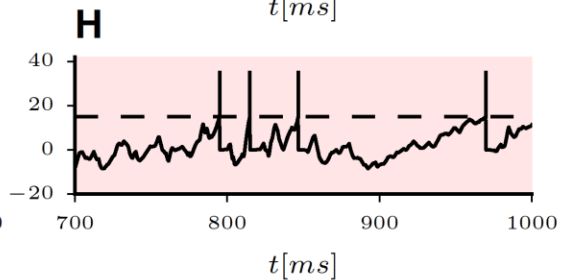
balanced



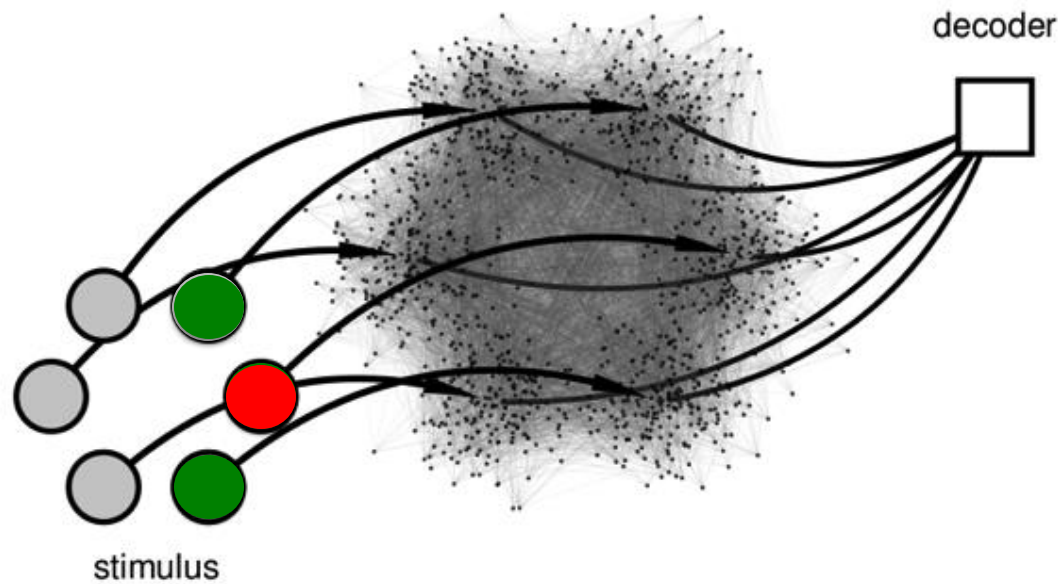
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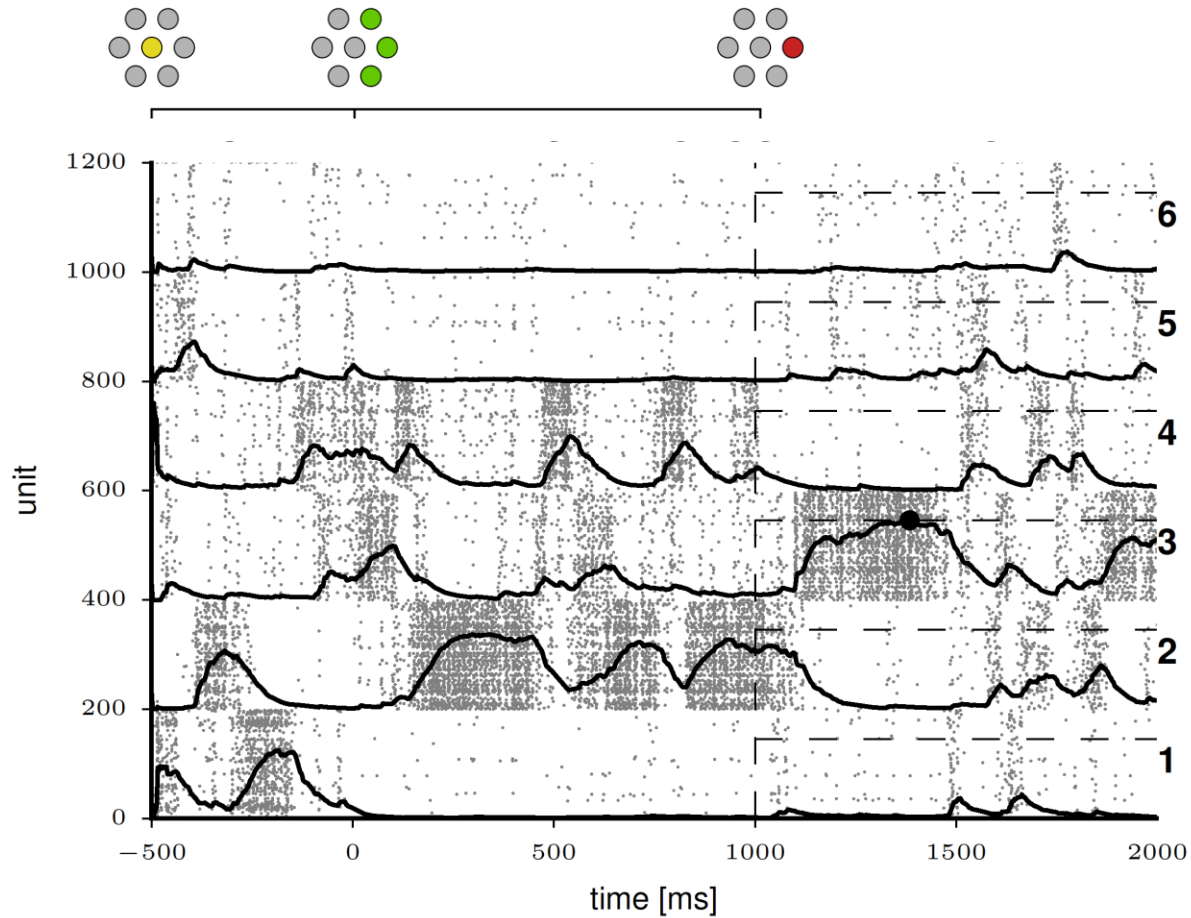
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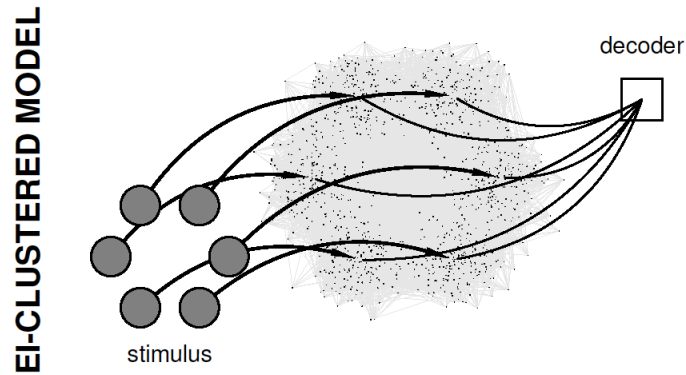
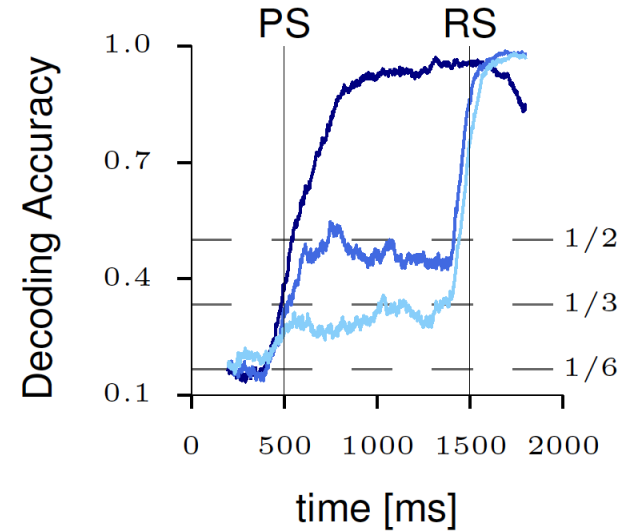
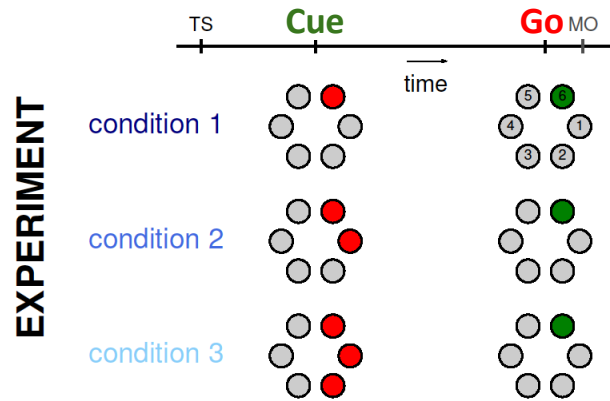
Spiking Attractor Network in Experimental Task Situation



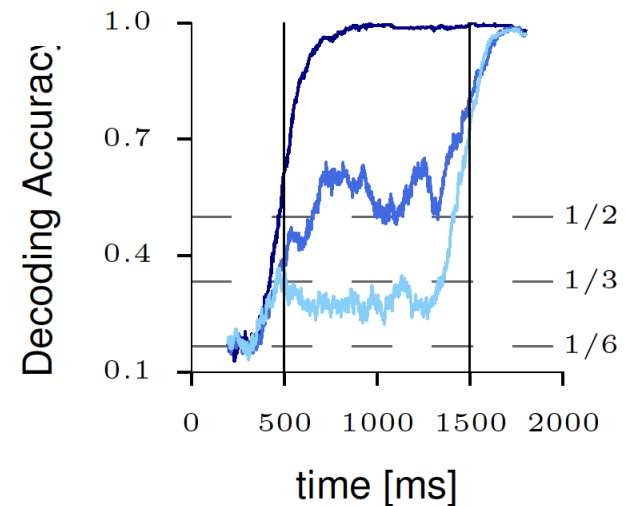
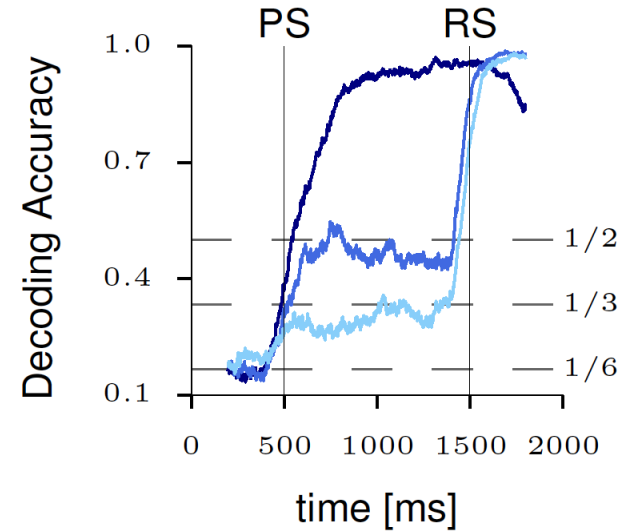
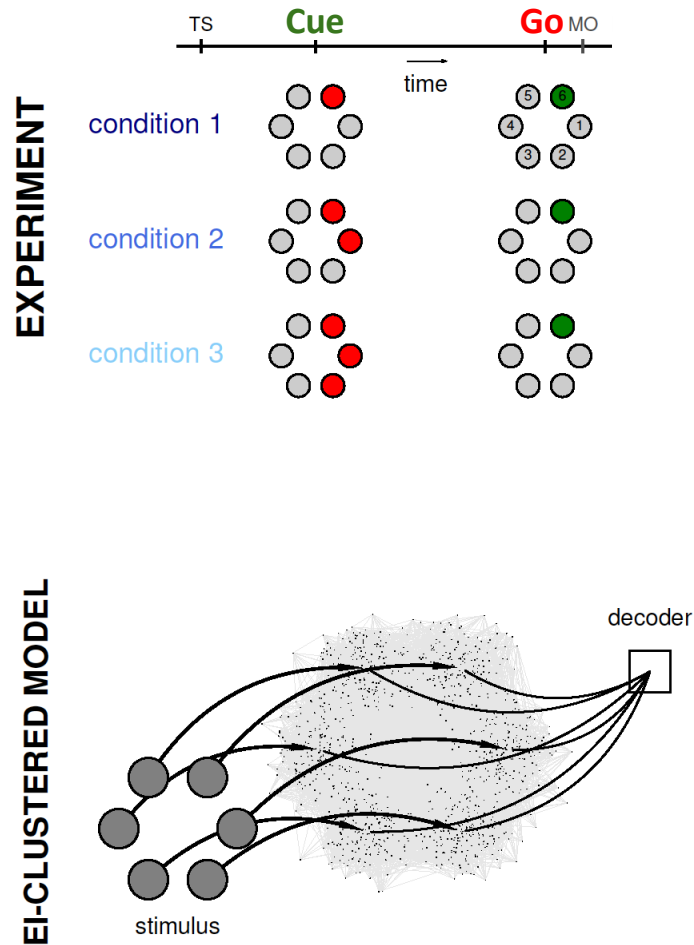
Single Trial Simulation and Movement Initiation



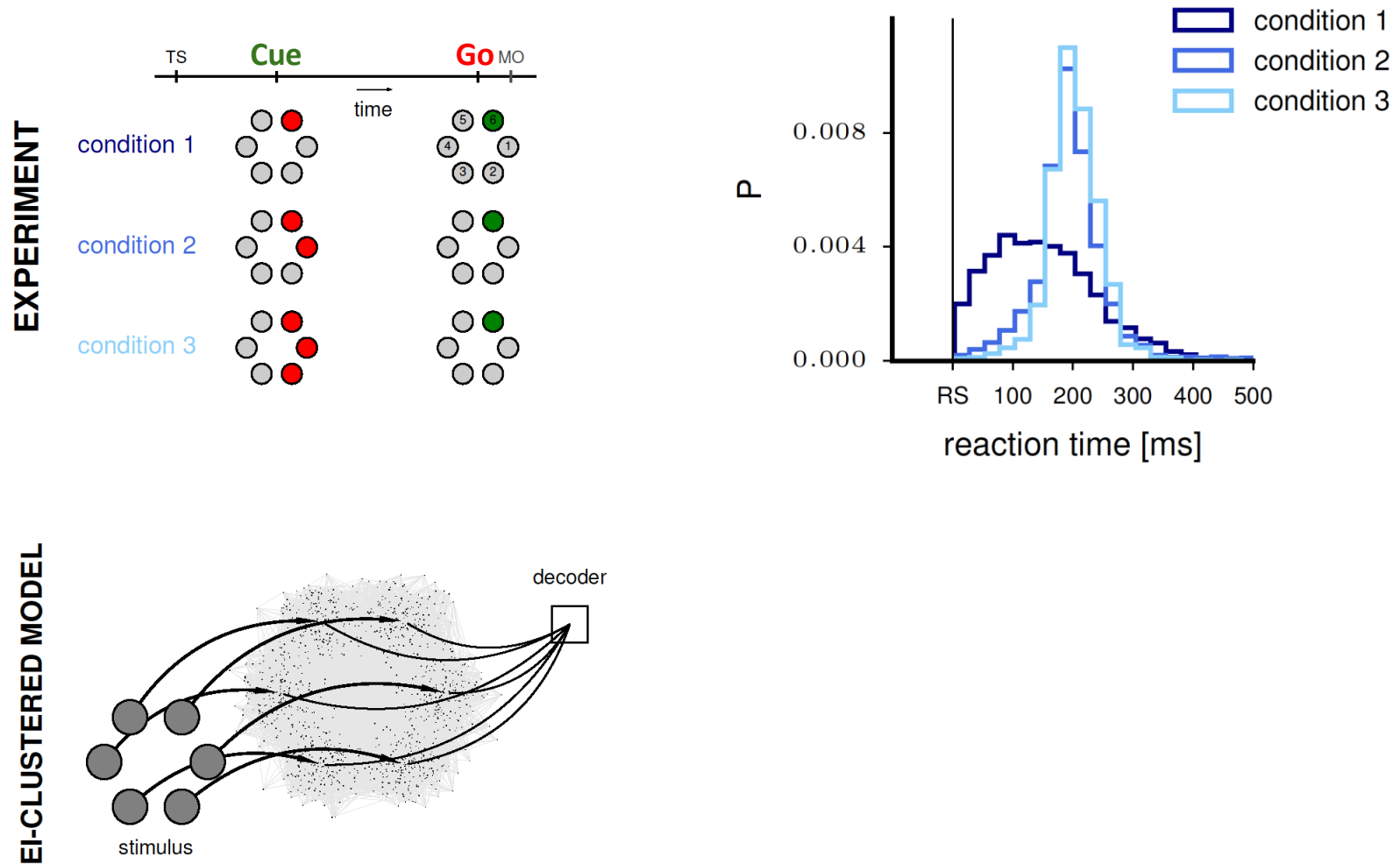
Task Uncertainty reflected in Directional Encoding *In Vivo* and *In Silico*



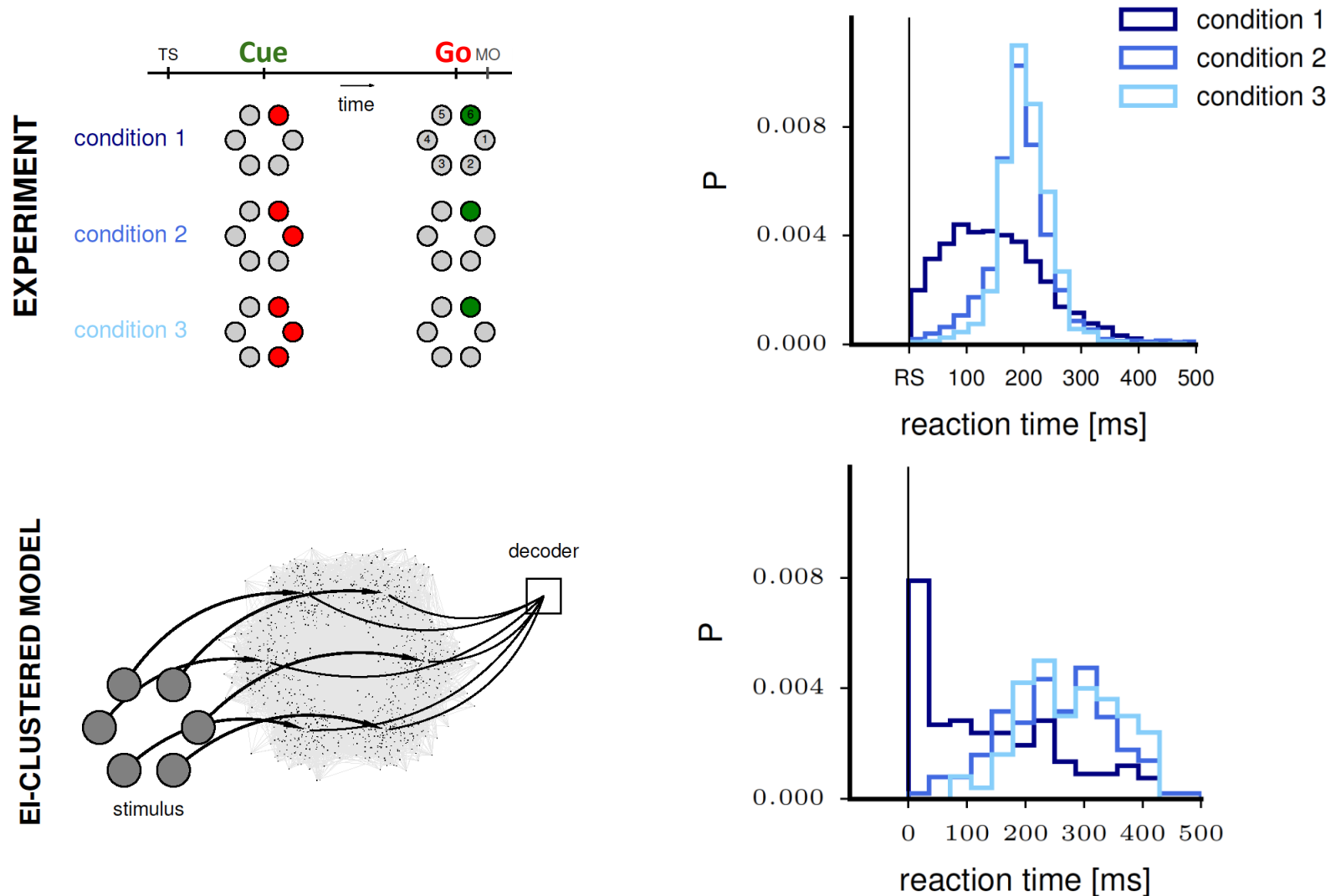
Task Uncertainty reflected in Directional Encoding *In Vivo* and *In Silico*



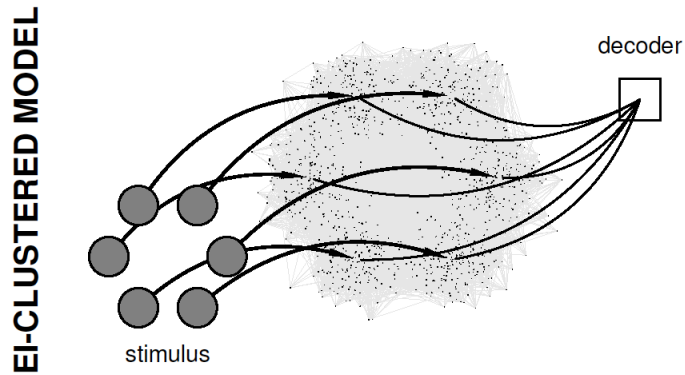
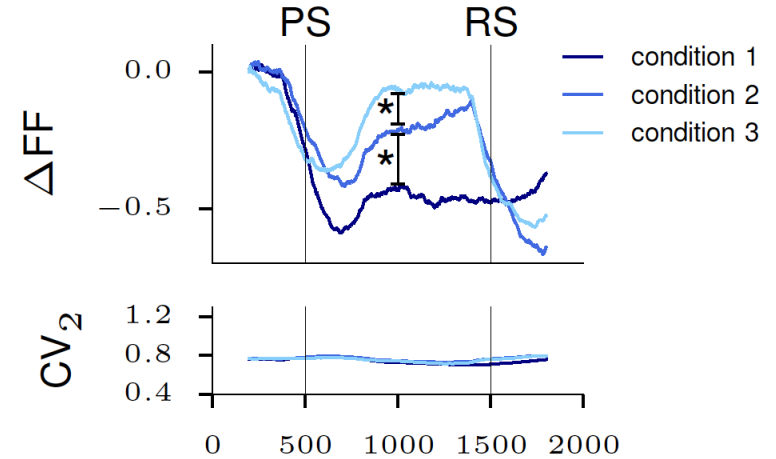
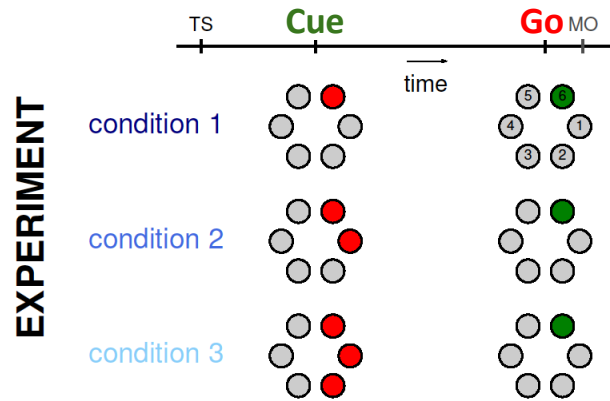
Task Uncertainty reflected in Reaction Times *In Vivo* and *In Silico*



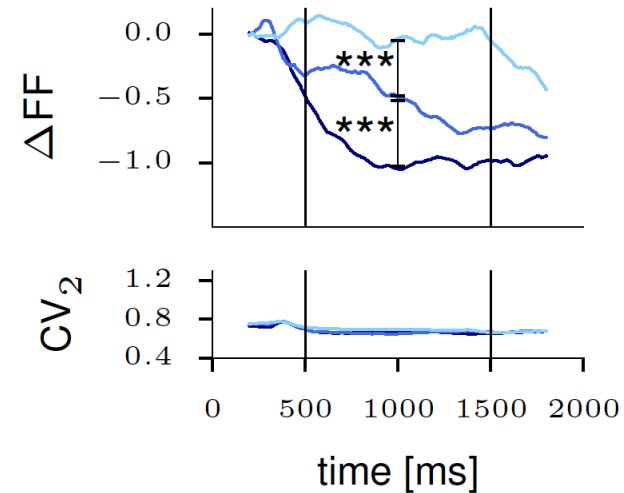
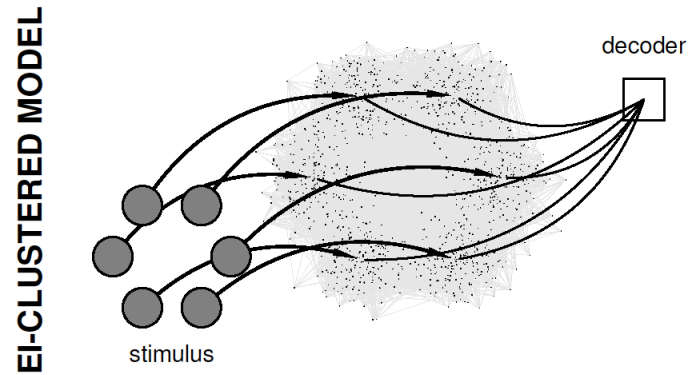
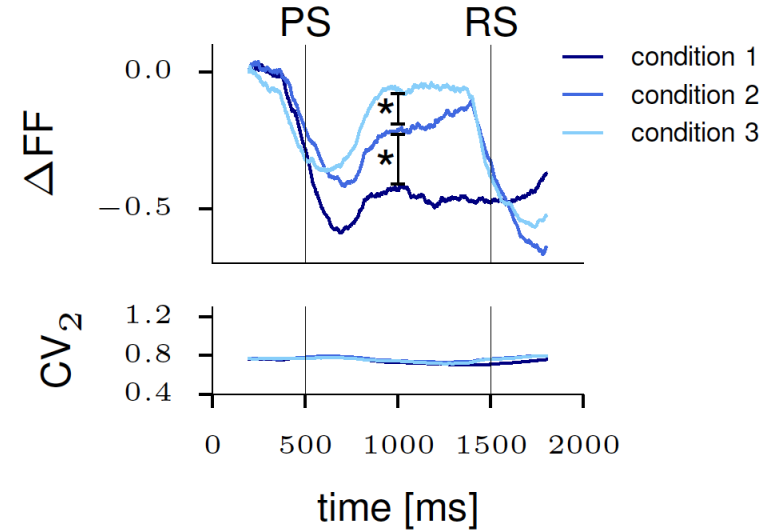
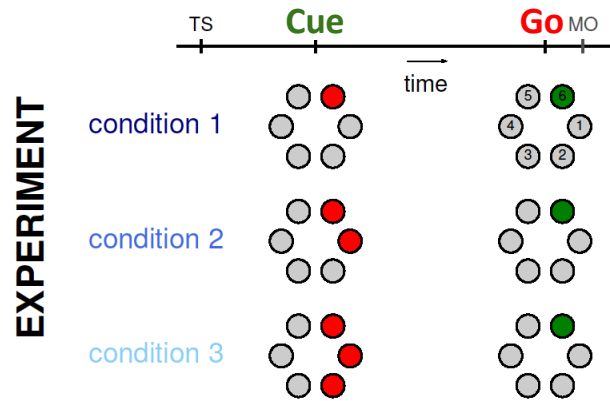
Task Uncertainty reflected in Reaction Times *In Vivo* and *In Silico*



Task Uncertainty reflected in Variability Dynamics *In Vivo* and *In Silico*

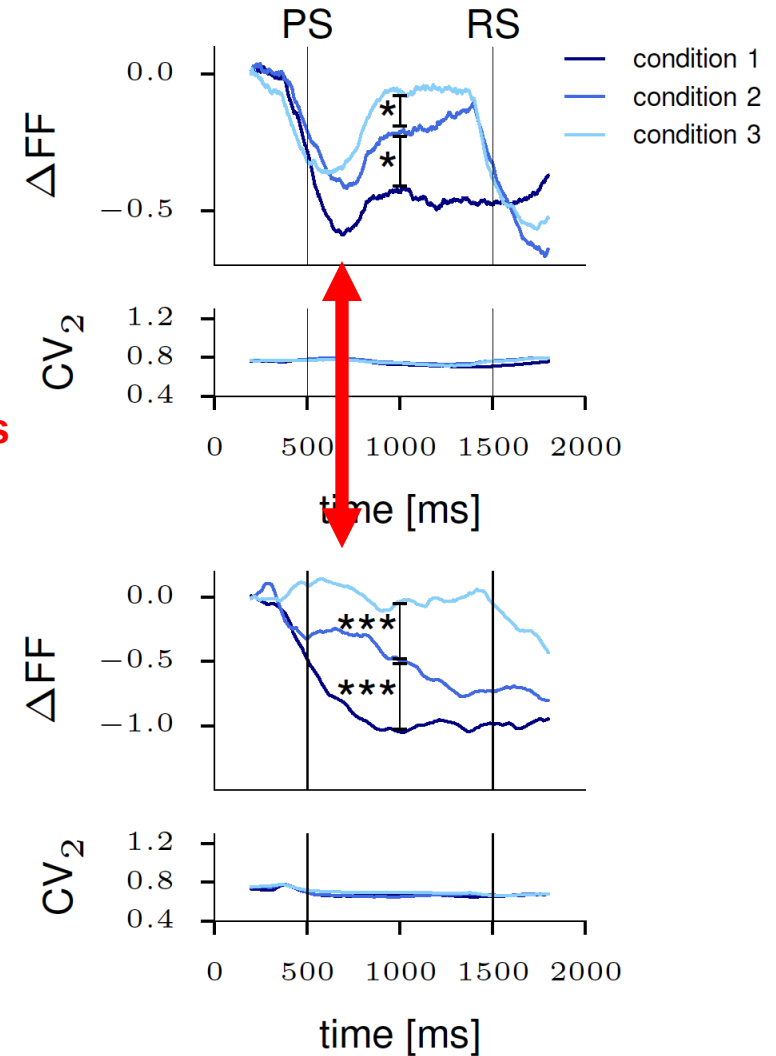


Task Uncertainty reflected in Variability Dynamics *In Vivo* and *In Silico*



Model 2c: Excitatory-Inhibitory (E/I) Cluster Topology + Adaptation

missmatch in early FF response dynamics



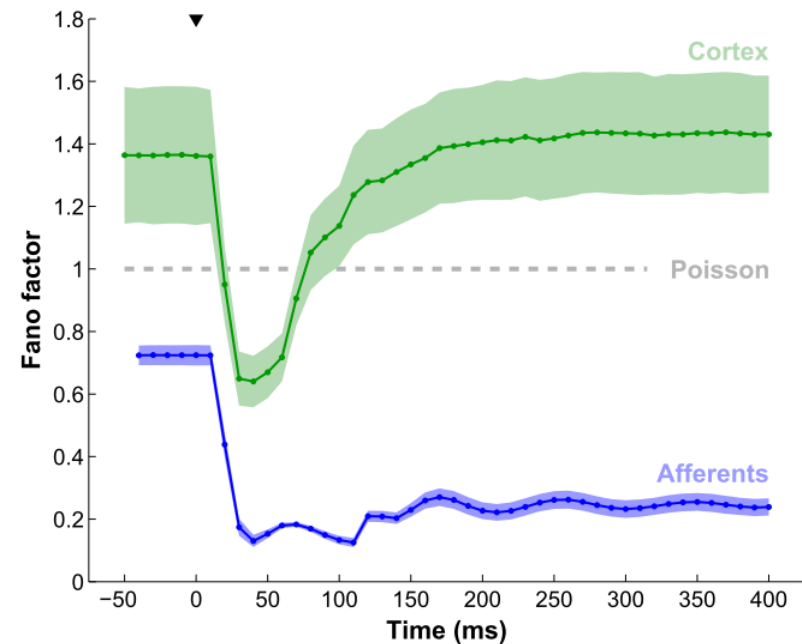
Model 2c: Excitatory-Inhibitory (E/I) Cluster Topology + Adaptation

$$c_m \frac{dv(t)}{dt} = g_l(E_l - v(t)) + g_e(t)(E_e - v(t)) + g_i(t)(E_i - v(t)) \\ + g_s(t)(E_s - v(t))$$

If v exceeds the threshold v_{th} :

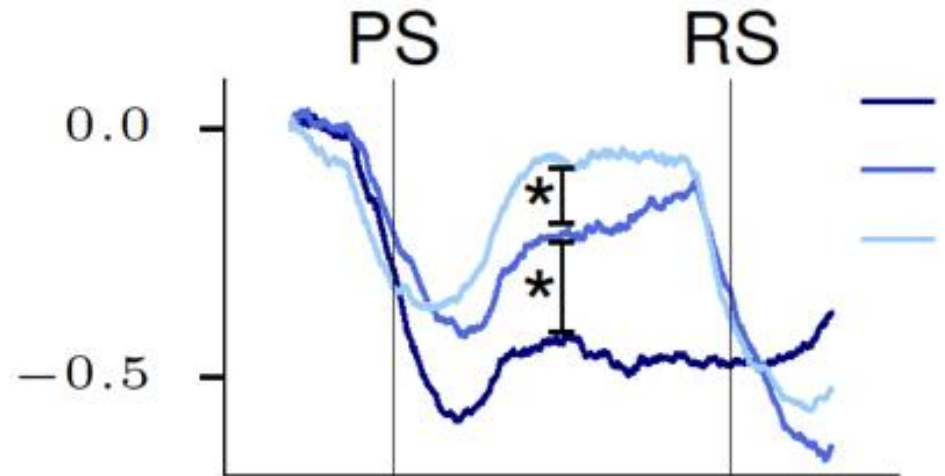
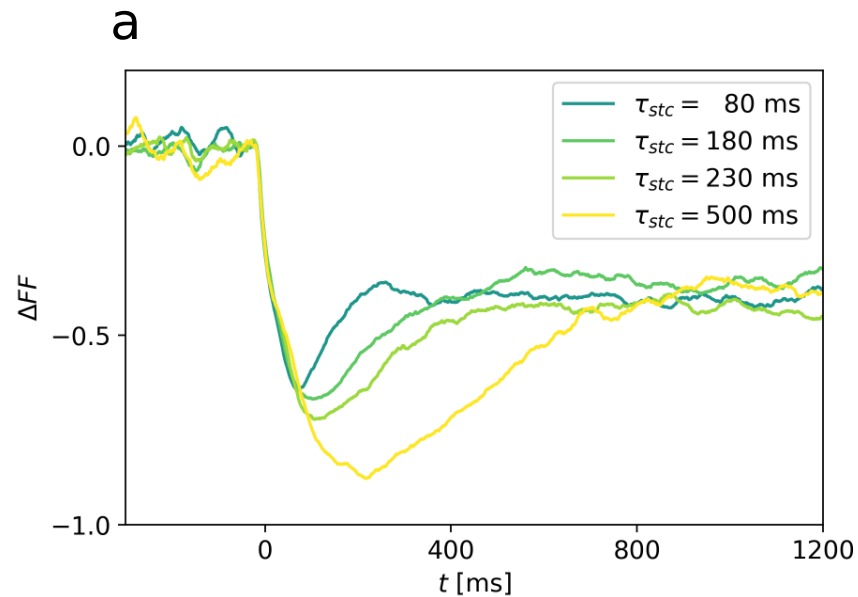
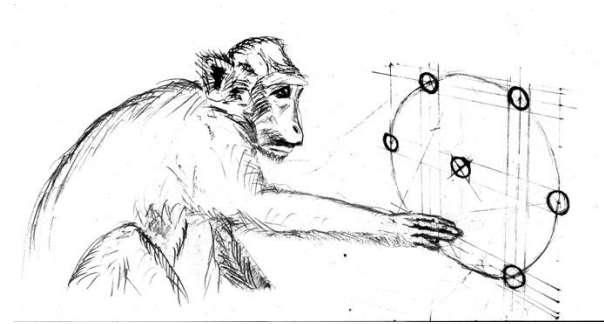
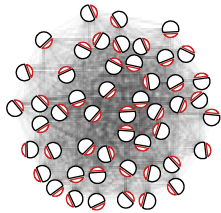
- v is reset to v_{reset} .
- $g_s \mapsto g_s + q_s$.

slow SFA conductance ($\tau_s > \tau_m$)



Model 2c: Excitatory-Inhibitory (E/I) Cluster Topology + Adaptation

E/I - clustering + SFA



Felix Schmitt, MS in preparation

Take Home's - Part II

SOURCES OF VARIABILITY

- Neuron intrinsic factors (synaptic transmission, integration, spiking) make a **negligible contribution** to the output variability
- **Balanced network input** can explain **<50%** of output variability ($FF \leq CV^2$)
- Excess output variability *in vivo* ($FF \gg CV^2$) reflects non-stationarity across trials that can be caused by
 - **slow and weak modulation** of the network state
 - **attractor dynamics** in clustered networks

SINKS OF VARIABILITY

- **Spike Frequency Adaptation**
 - **reduces variability** at single neuron and population level
 - introduces negative serial interval correlations

THANKS (a.u.)

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